



University of Idaho
College of Science



**Systems Medicine
of Infectious Diseases**
Research and Consulting Group

Adaptive Observers for Medical Digital Twins

A Framework for personalized medicine

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Department of Mathematics and Statistical Science
University of Idaho

Panel 1. A 52-year-old male with a history of stroke and hemiplegia is found confused in his prison cell.



Taken, and adapted with Gemini, from:

[1] Laubenbacher, R., Mehrad, B., Shmulevich, I., & Trayanova, N. (2024). **Digital twins in medicine.** Nature Computational Science, 4(3), 184-191.

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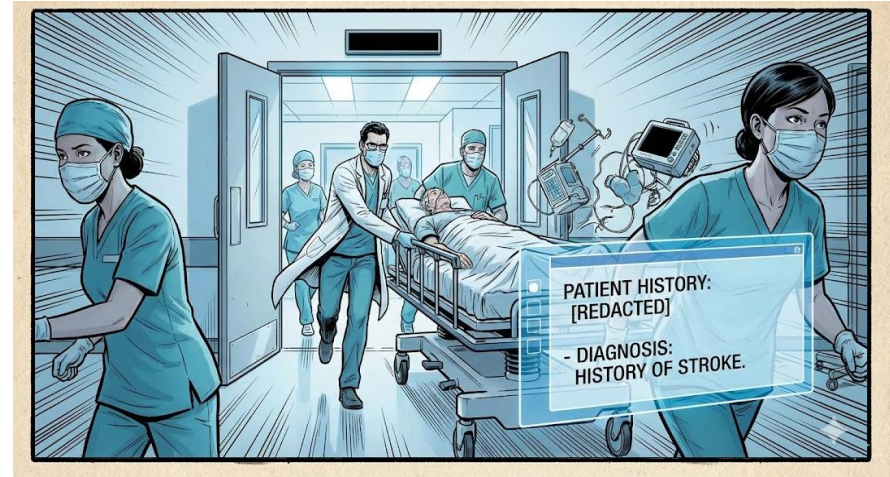
Panel 2. At ED, patient presents hypotension, mild renal insufficiency, and bilateral airspace disease on chest X-ray.



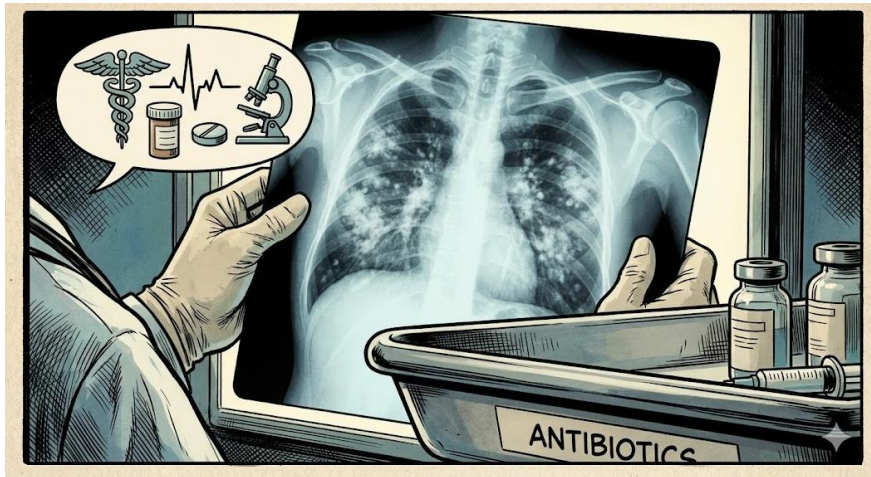
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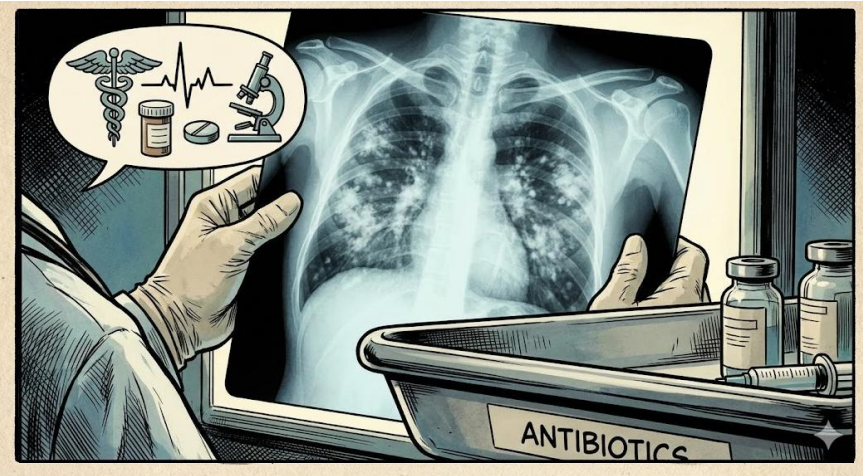
Panel 3. Pneumonia diagnosed and empiric severe pathogen antibiotics started.



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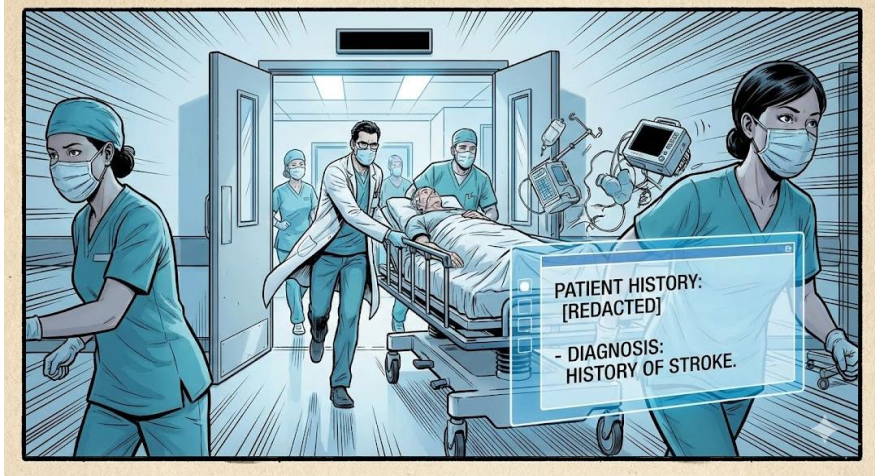


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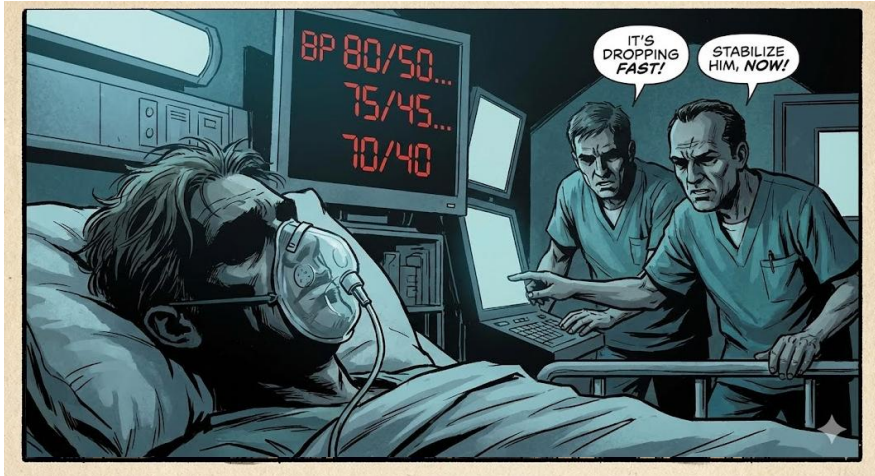


Suspected pathogen: ***Klebsiella pneumoniae***.

Panel 2. At ED, patient presents hypotension, mild renal insufficiency, and bilateral airspace disease on chest X-ray.



Panel 4. 6 h later, recurrent hypotension requiring intravenous pressors and escalating supplemental oxygen.



Panel 5. 3 h after deterioration, requires endotracheal intubation and mechanical ventilation for ARDS-induced hypoxic respiratory failure.



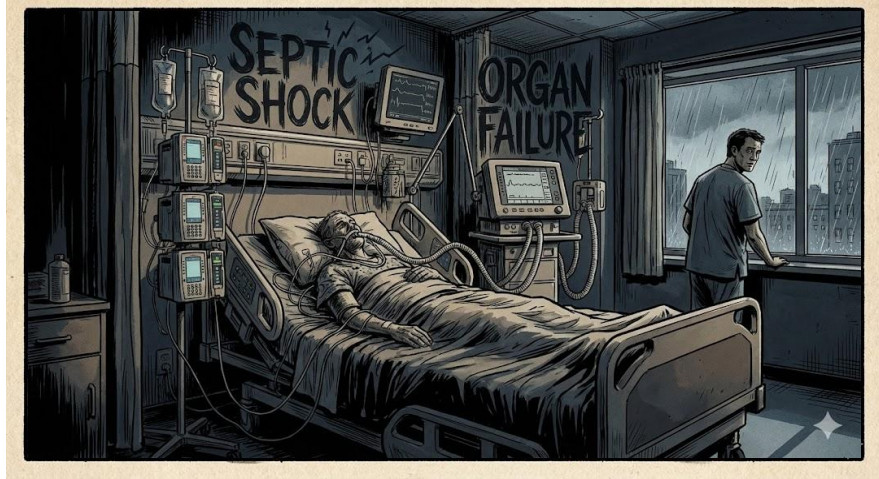
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Panel 6. Condition progress to refractory septic shock and acute kidney injury **despite receiving antibiotics to which the *Klebsiella pneumoniae* was sensitive.**



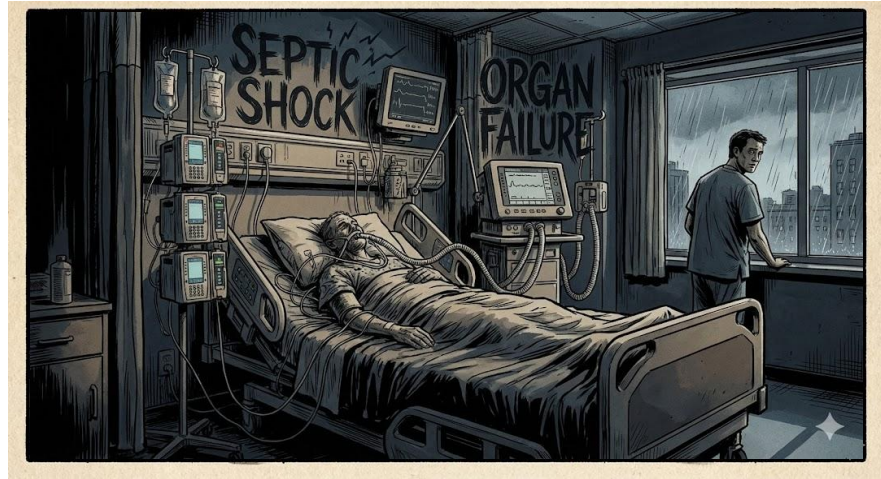
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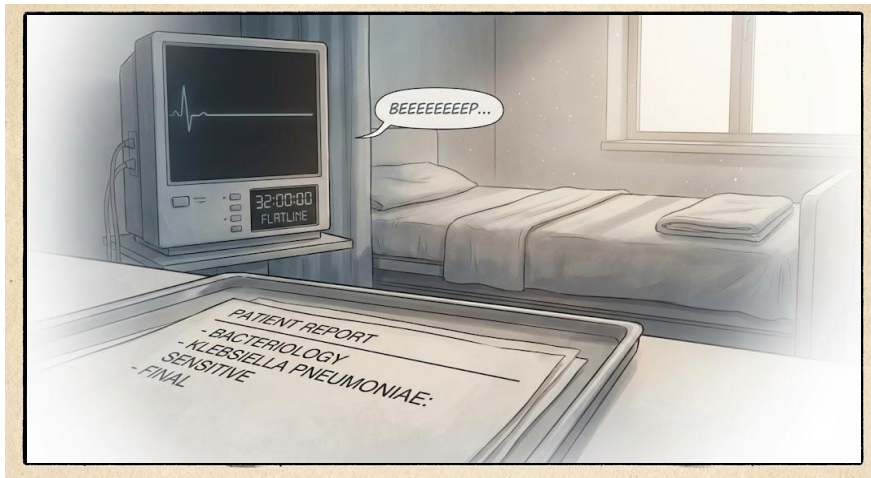
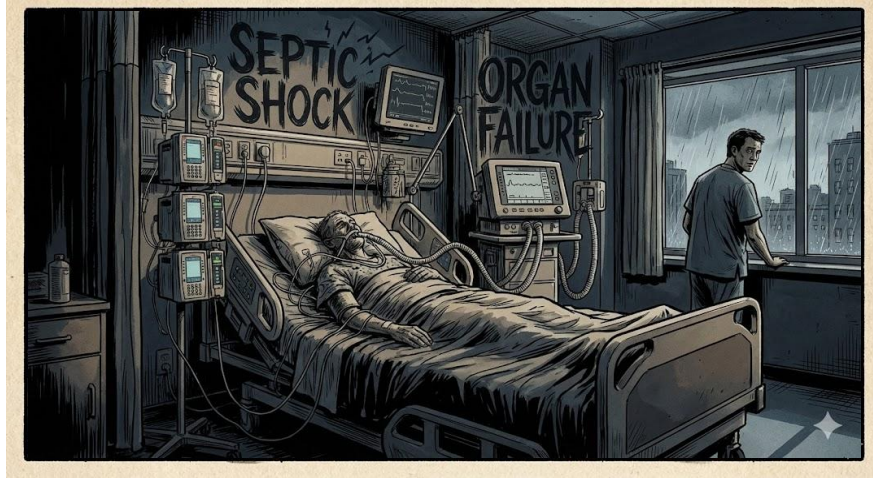
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Panel 7. Patient dies of multi-organ failure 32 hours after presentation.

Why we need **Digital Twins in medicine?**
This one represent the biggest promise and challenge in applying computational modeling at the level of individual patients, i.e., personalized medicine.

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What is a Digital Twin in Medicine?

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Digital Twin [2]:

1. A model of the physical twin.
2. An evolving set of data relating to the object, and
3. A means of dynamically updating or adjusting the model in accordance with data.

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NASA reports [3]:

A **digital twin** is an integrated multiphysics, multi-scale, probabilistic simulation of a vehicle or system that uses the best available physical models, sensor updates, flight history, etc., to mirror the life of its flying twin. The **digital twin** is ultra-realistic and may consider one or more important and interdependent vehicle systems, including propulsion and energy storage, life support, avionics, thermal protection, etc.

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A trade-off arises

What is a Digital Twin in Medicine?

Digital Twin:

1. A model of the physical twin.
2. An evolving set of data relating to the object, and
3. A means of dynamically updating or adjusting the model in accordance with data.

Medical Digital Twin [1,4]:

A computational model of the system to be twinned (heathy or sickness processes) that is connected to the physical twin in a bidirectional fashion over time, periodically recalibrated with patient data, and provides patient predictions over time to prevent or treat diseases.

Taken from:

[1] Laubenbacher, R., Mehrad, B., Shmulevich, I., & Trayanova, N. (2024). **Digital twins in medicine**. Nature Computational Science, 4(3), 184-191.

[4] Foundational Research Gaps and Future Directions for Digital Twins (National Academies of Sciences, Engineering and Medicine, 2023); <https://doi.org/10.17226/26894>.

Digital Twins in Medicine

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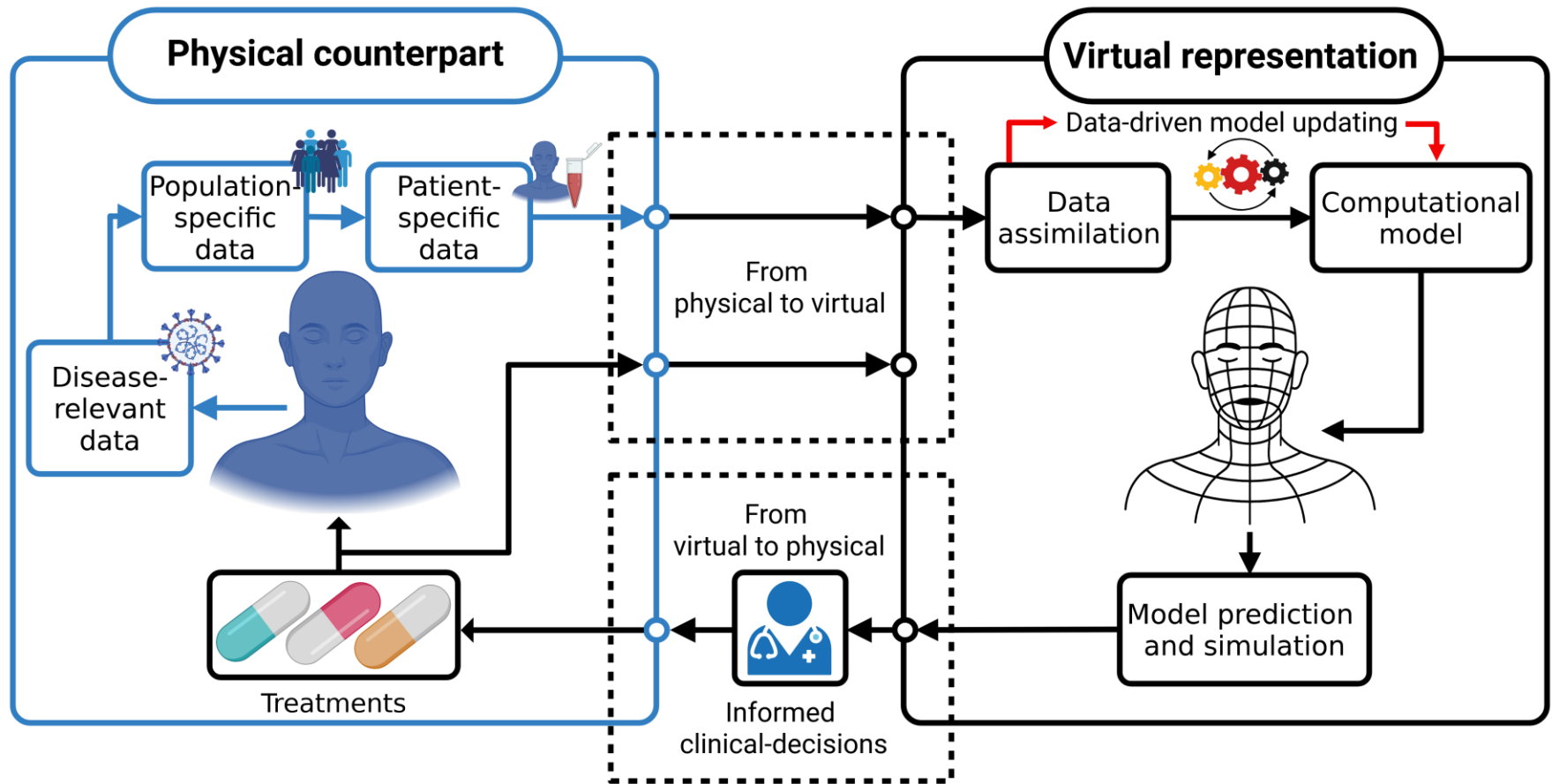


Figure 1.a. The standard Digital Twin Framework [5] .

Digital Twins in Medicine

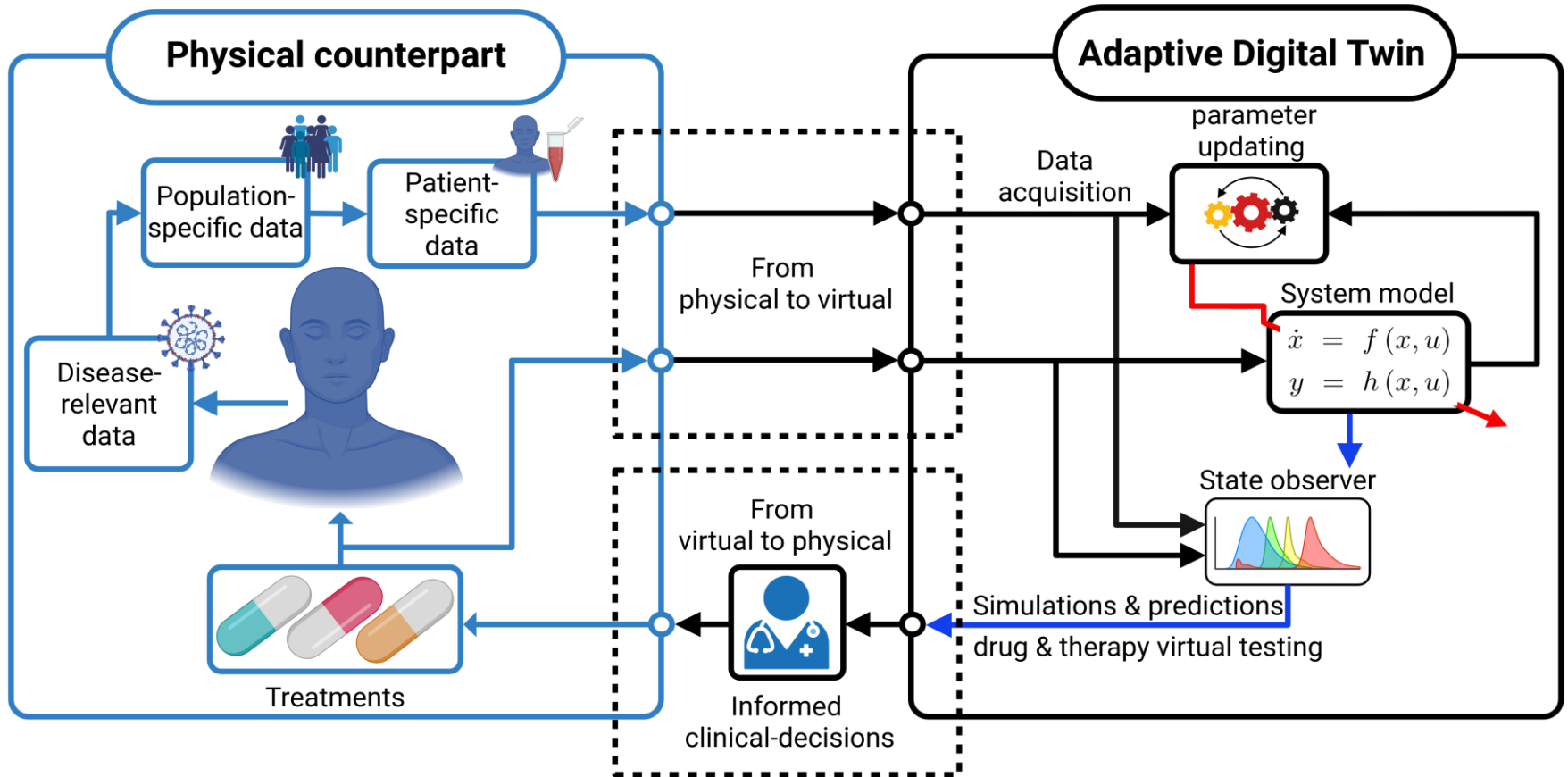


Figure 1.b. Our Adaptive Digital Twin Framework [5] .

Mathematical foundations for model parameter updating

Model Parameter Updating:

1. Adaptive Observers [6].
2. Hybrid Extended Kalman Filter [7].

Taken from:

[6] Besancon, G., de León-Morales, J., & Huerta-Guevara, O. (2006). *On adaptive observers for state affine systems*. International journal of Control, 79(06), 581-591.

[7] Simon, D. (2006). *Optimal state estimation: Kalman, H infinity, and nonlinear approaches*. John Wiley & Sons.

Mathematical foundations for model parameter updating

Model Parameter Updating:

1. Adaptive Observers [6].
2. Hybrid Extended Kalman Filter [7].

Biomedical applications:

1. HIV mutation model [8].
2. Antimicrobial Resistance model [9] .

Taken from:

[8] Hernandez-Vargas, E. A., & Middleton, R. H. (2013). ***Modeling the three stages in HIV infection***. Journal of theoretical biology, 320, 33-40.

[9] Hernández-Vargas, E. A., Parra-Rojas, C., & Oлару, S. (2021, December). ***Switching logistic maps to design cycling approaches against antimicrobial resistance***. In 2021 60th IEEE conference on decision and control (CDC) (pp. 4248-4253). IEEE.

HIV mutation model

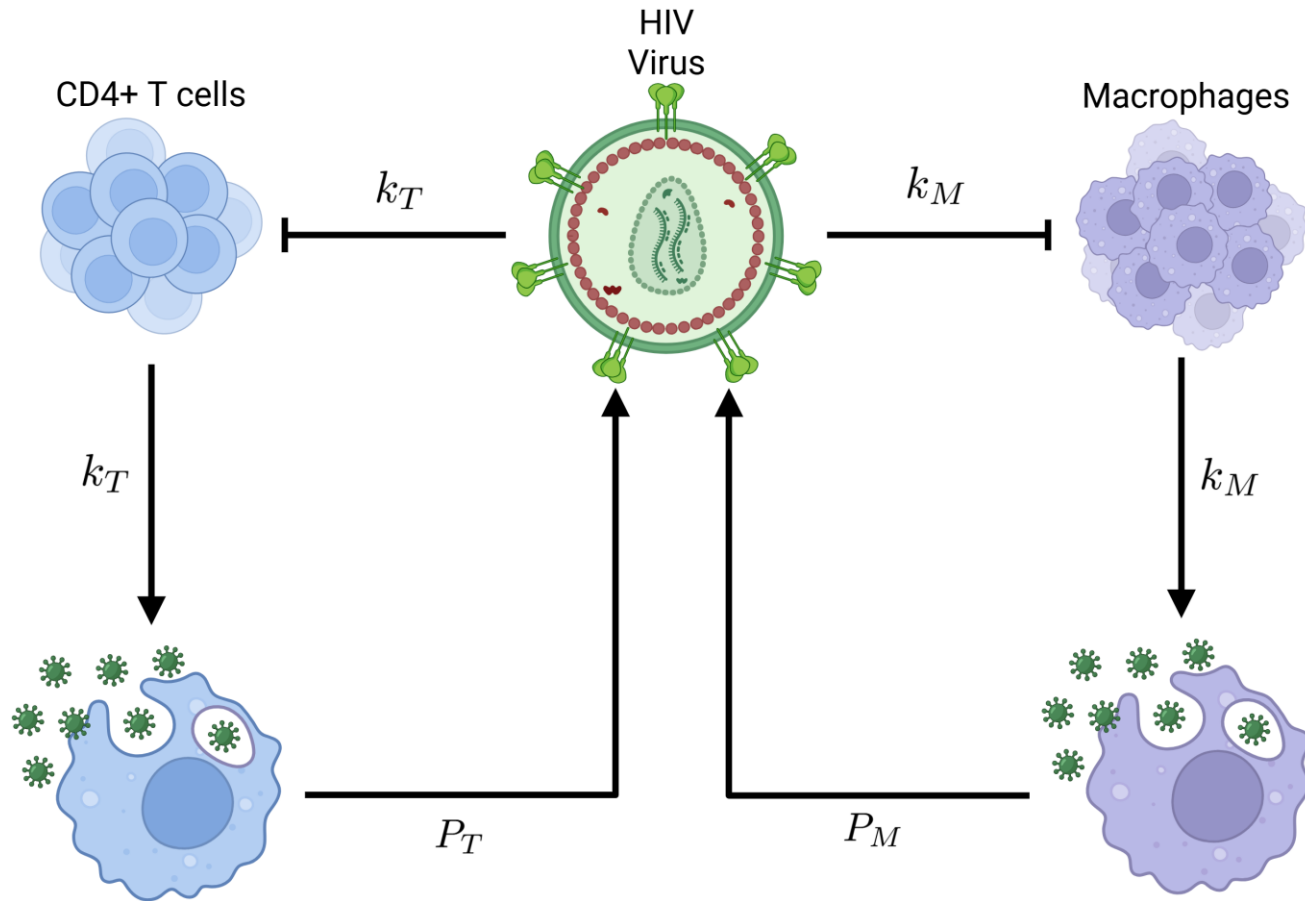


Figure 2. HIV infection scheme, taken from [8].

HIV mutation model

$$\dot{T} = s_T + \frac{\rho_T}{C_T + V_T} T V_T - \sum_{i=1}^n k_{T,\sigma}^i T V_i - \delta_T T$$

$$\dot{M} = s_M + \frac{\rho_M}{C_M + V_T} M V_T - \sum_{i=1}^n k_{M,\sigma}^i M V_i - \delta_M M$$

$$\dot{T}_i^* = k_{T,\sigma}^i T V_i + \sum_{j=1}^n \mu m_{i,j} V_j T - \delta_{T^*} T_i^*$$

$$\dot{M}_i^* = k_{M,\sigma}^i M V_i + \sum_{j=1}^n \mu m_{i,j} V_j M - \delta_{M^*} M_i^*$$

$$\dot{V}_i = p_{T,\sigma}^i T_i^* + p_{M,\sigma}^i M_i^* - \delta_V V_i.$$

HIV mutation model

$$\dot{T} = s_T + \frac{\rho_T}{C_T + V_T} T V_T - \sum_{i=1}^n k_{T,\sigma}^i T V_i - \delta_T T$$

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$$\dot{T}_i^* = k_{T,\sigma}^i T V_i + \sum_{j=1}^n \mu m_{i,j} V_j T - \delta_{T^*} T_i^*$$

$$\dot{M}_i^* = k_{M,\sigma}^i M V_i + \sum_{j=1}^n \mu m_{i,j} V_j M - \delta_{M^*} M_i^*$$

$$\dot{V}_i = p_{T,\sigma}^i T_i^* + p_{M,\sigma}^i M_i^* - \delta_V V_i. \quad (1)$$

Can we update model parameters to estimate infections rates and the number of infected cells from measurements?

Adaptive Observers

Adaptive Observers

An **adaptive observer** is a type of observer used in adaptive control theory that allow to estimate the states variables while adapting unknown parameters of the system.

Taken from:

[6] Besancon, G., de León-Morales, J., & Huerta-Guevara, O. (2006). ***On adaptive observers for state affine systems***. International journal of Control, 79(06), 581-591.

Adaptive Observers

An **adaptive observer** is a type of observer used in adaptive control theory that allow to estimate the states variables while adapting unknown parameters of the system.

Its main advantage is that the estimation of states and parameters is **decoupled**. This Approach assumes that the former has faster dynamics than the latter.

Taken from:

[6] Besancon, G., de León-Morales, J., & Huerta-Guevara, O. (2006). ***On adaptive observers for state affine systems***. International journal of Control, 79(06), 581-591.

Adaptive Observers

Consider a proper change of coordinates that allow to rewritten model (1) in the following state affine form:

$$\begin{aligned}\dot{x} &= A(y)x + \beta(y) + \varphi(y)\theta, \\ y &= Cx,\end{aligned}\tag{2}$$

where $x \in \mathbb{R}^{n_x}$, $y \in \mathbb{R}^{n_y}$, and $\theta \in \mathbb{R}^{n_\theta}$ are the state vector, measurable output vector and unknown parameter vector, respectively. Under assumption that y are measurable signal and matrices $A(y)$, $\beta(y)$ and $\varphi(y)$ are uniformly bounded continuous functions depending only on y , therefore the adaptive observer:

$$\begin{aligned}\dot{\hat{x}} &= A\hat{x} + \beta + \varphi\hat{\theta} + \{S^{-1}C^T \\ &\quad + \Lambda\Gamma^{-1}\Lambda^TC^T\}Q(y - C\hat{x}) \\ \dot{S} &= -\rho_x S - A^TS - SA + C^TQC \\ \dot{\Lambda} &= \{A - S^{-1}C^TQC\}\Lambda + \varphi \\ \dot{\Gamma} &= -\rho_\theta\Gamma + \Lambda^TC^TQC\Lambda \\ \dot{\hat{\theta}} &= \Gamma^{-1}\Lambda^TC^TQ(y - C\hat{x}),\end{aligned}\tag{3}$$

Taken from:

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Adaptive Observers

Consider the HIV model (1), the following change of coordinates

$$x = [T, M, T^*, M^*, V]/\lambda,$$

with,

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} x,$$

transforms the model (1) into

$$\begin{aligned} \dot{x}_1 &= -\delta_T x_1 + \left(\frac{\lambda \rho_T y_1 y_2}{C_T + \lambda y_2} + \frac{S_T}{\lambda} \right) - \lambda y_1 y_2 K_{T,\sigma}^1, \\ \dot{x}_2 &= \left(\frac{\lambda \rho_M y_2}{C_M + \lambda y_2} - \lambda y_2 K_{M,\sigma}^1 - \delta_M \right) x_2 + \frac{S_M}{\lambda}, \\ \dot{x}_3 &= -\delta_{T^*} x_3 + \lambda y_1 y_2 K_{T,\sigma}^1, \\ \dot{x}_4 &= \lambda y_2 K_{M,\sigma}^1 x_2 - \delta_{M^*} x_4, \\ \dot{x}_5 &= P_{T,\sigma} x_3 + P_{M,\sigma} x_4 - \delta_V x_5, \end{aligned}$$

where $\lambda \in \mathbb{R}$ is a known constant scaling factor arbitrarily chosen for convenience of numerical computation.

Adaptive Observers

$$A = \begin{bmatrix} -\delta_T & 0 & 0 & 0 & 0 \\ 0 & \left(\frac{\lambda \rho_M y_2}{C_M + \lambda y_2} - \lambda y_2 K_{M,\sigma}^1 - \delta_M \right) & 0 & 0 & 0 \\ 0 & 0 & -\delta_{T^*} & 0 & 0 \\ 0 & \lambda y_2 K_{M,\sigma}^1 & 0 & -\delta_{M^*} & 0 \\ 0 & 0 & P_{T,\sigma} & P_{M,\sigma} & -\delta_V \end{bmatrix},$$

$$\beta = \begin{bmatrix} \frac{\lambda \rho_T y_1 y_2}{C_T + \lambda y_2} + \frac{S_T}{\lambda} \\ \frac{S_M}{\lambda} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \varphi = \begin{bmatrix} -\lambda y_1 y_2 \\ 0 \\ \lambda y_1 y_2 \\ 0 \\ 0 \end{bmatrix},$$

and finally

$$\theta = [K_{T,\sigma}^1],$$

Results on Real-time parameter updating

Numerical Results

Scenario 1.1: 1 genotype – 1 parameter

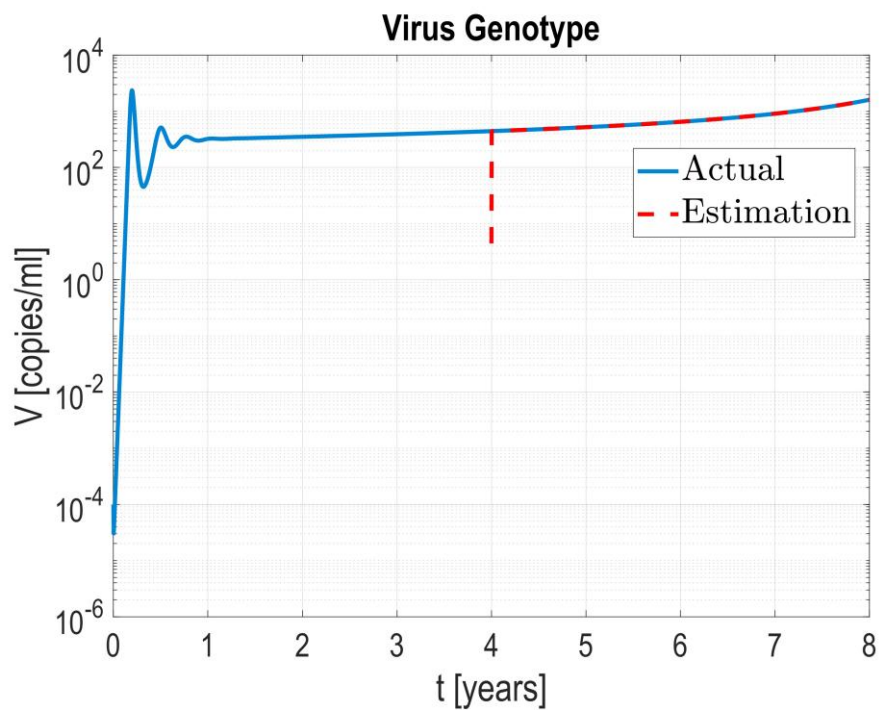


Figure 3. State estimation of Viral load

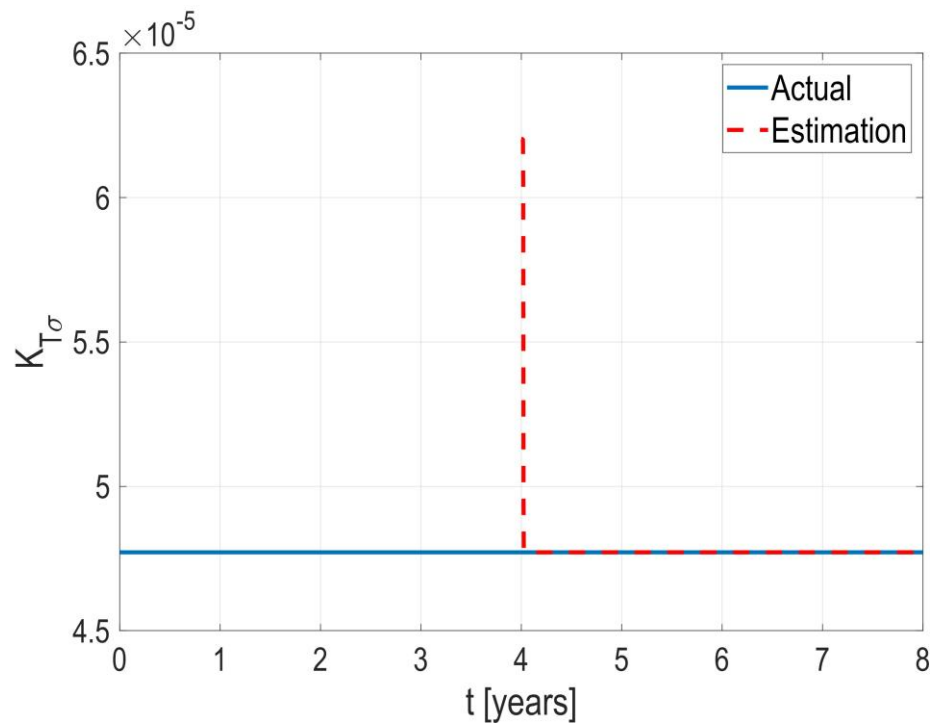


Figure 4. Parameter estimation of $K_{T,\sigma}$

Numerical Results

Scenario 1.1: 1 genotype – 1 parameter

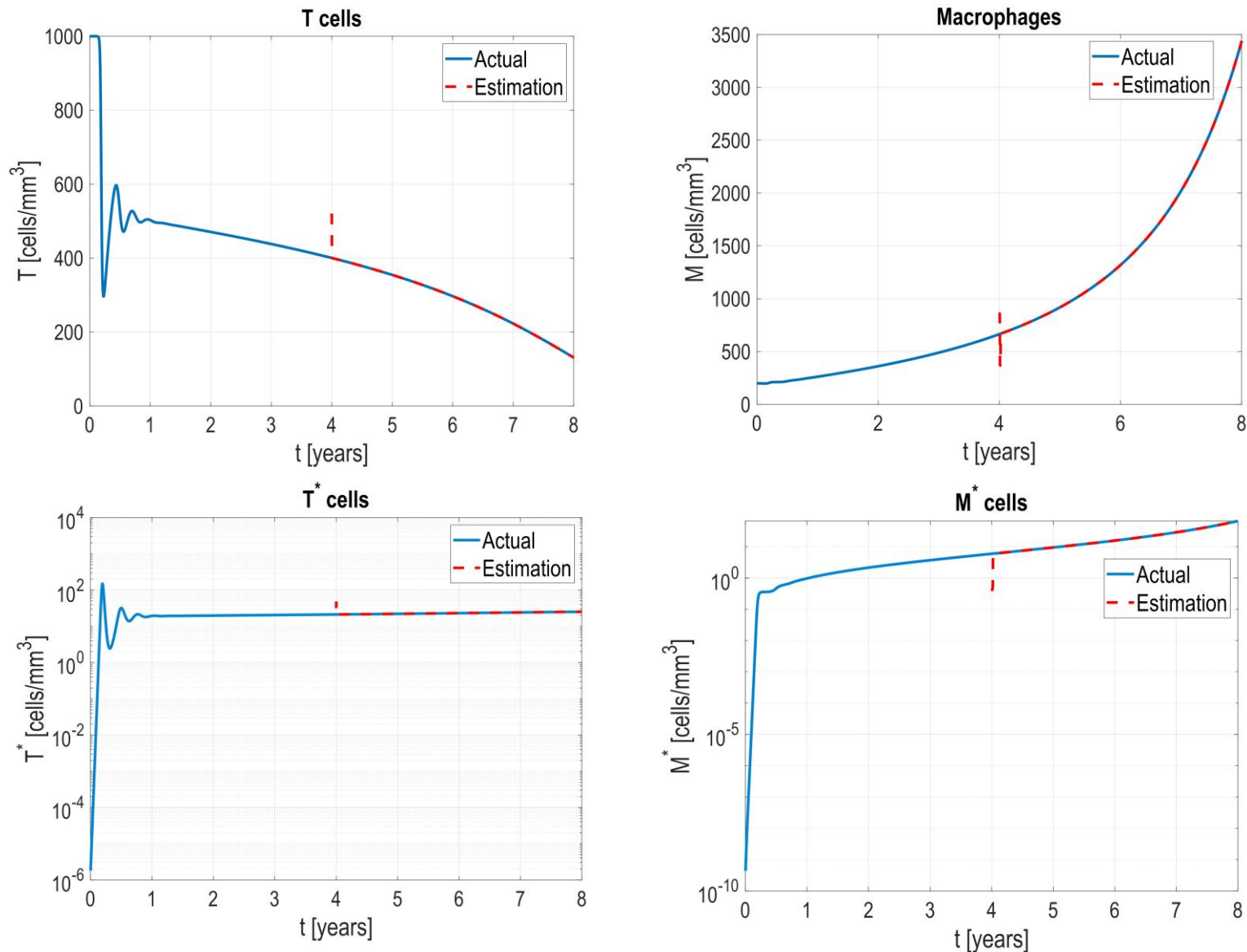


Figure 5. State estimation of HIV model

Numerical Results

Scenario 1.2: Switching between therapies.

Numerical Results

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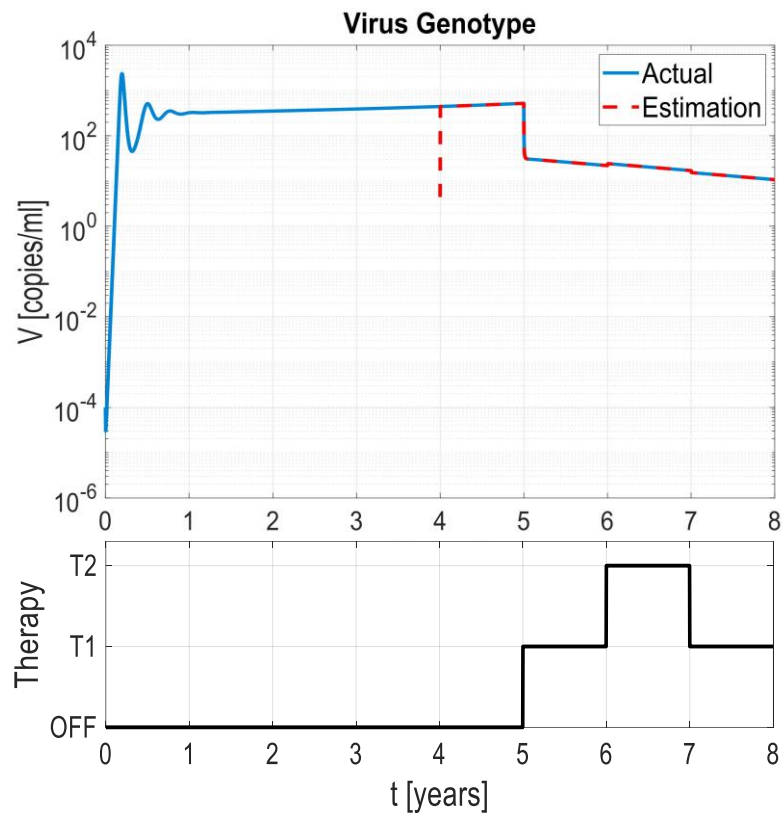


Figure 6. State estimation of Viral load

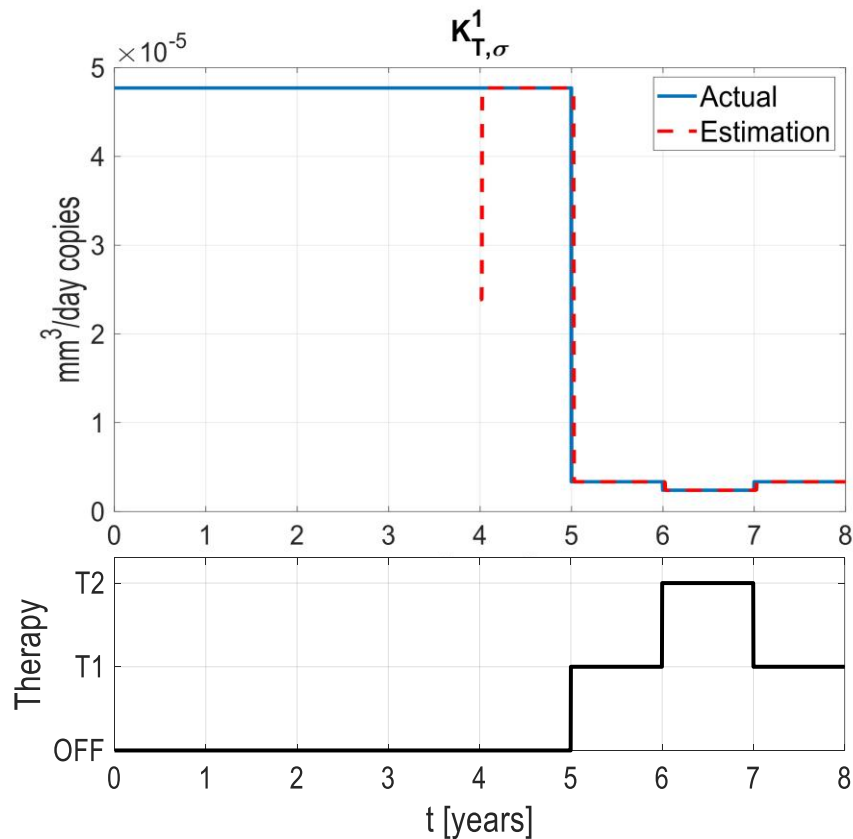


Figure 7. Parameter estimation of $K_{T,\sigma}$

Numerical Results

Scenario 1.2: Switching between therapies.

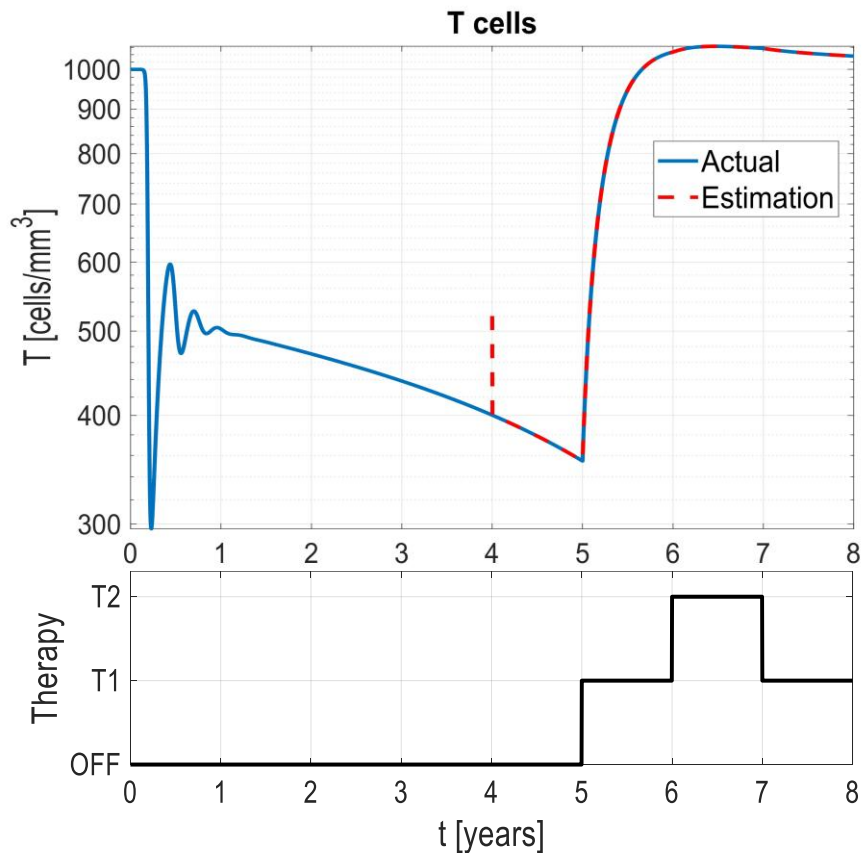


Figure 8. State estimation of CD4+ T cells

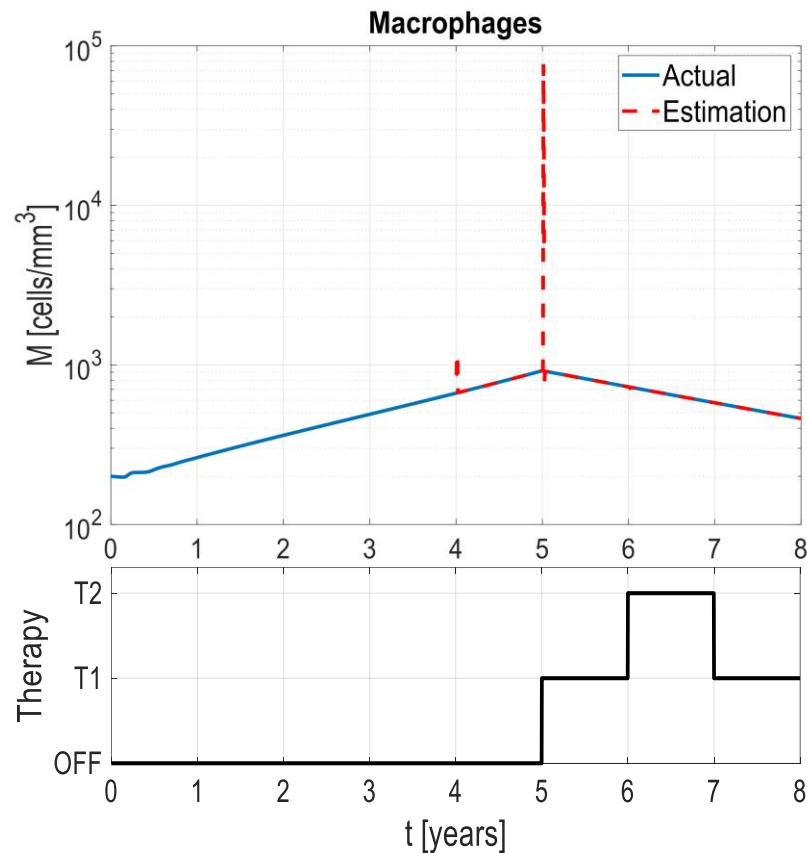


Figure 9. State estimation of Macrophages

Numerical Results

Scenario 1.2: Switching between therapies.

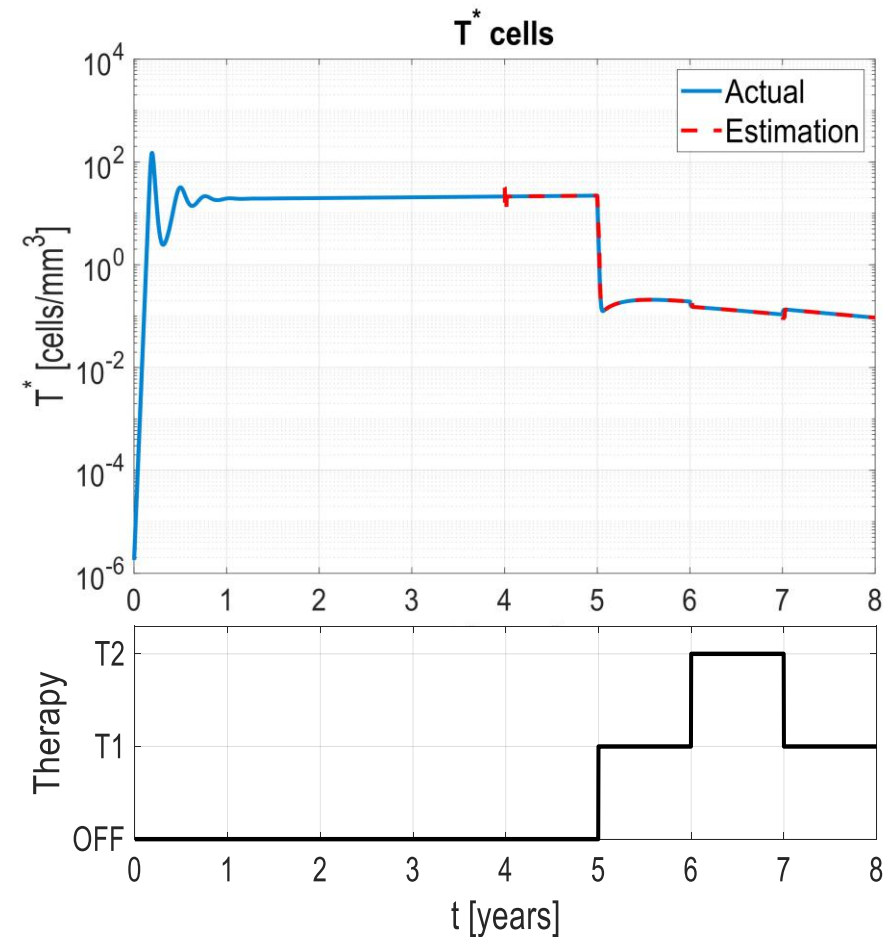


Figure 10. State estimation of Infected T cells

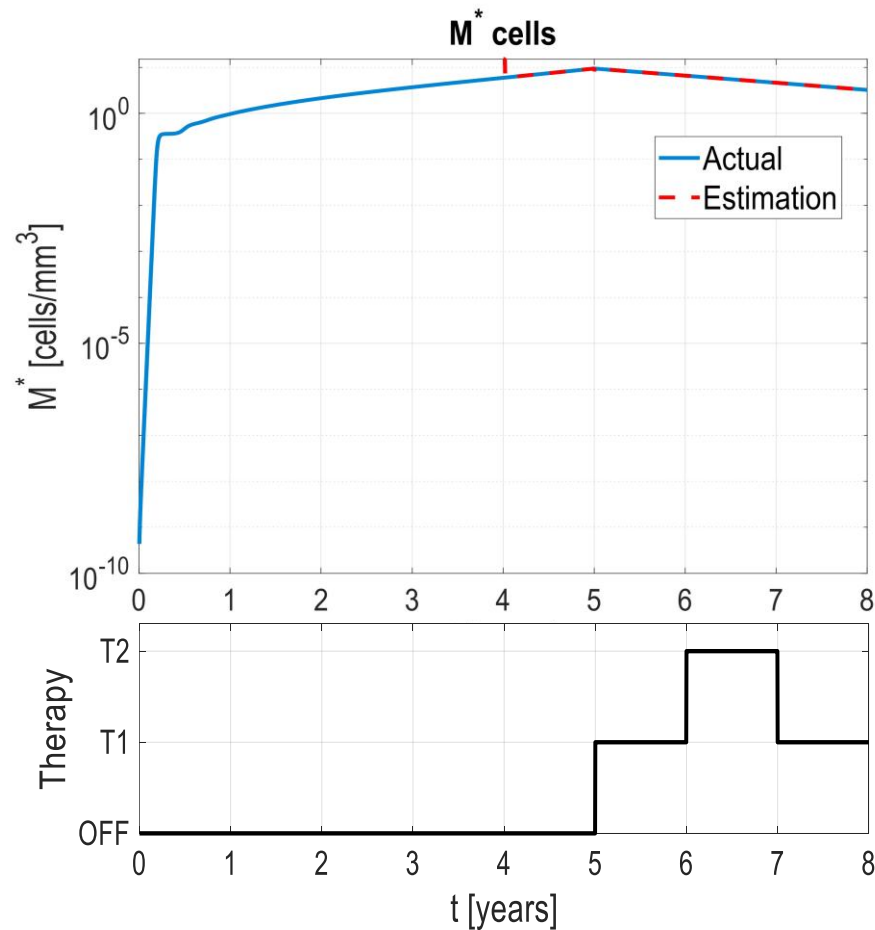


Figure 11. State estimation of Infected macrophages

HIV mutation

4 genotype – 4 parameter

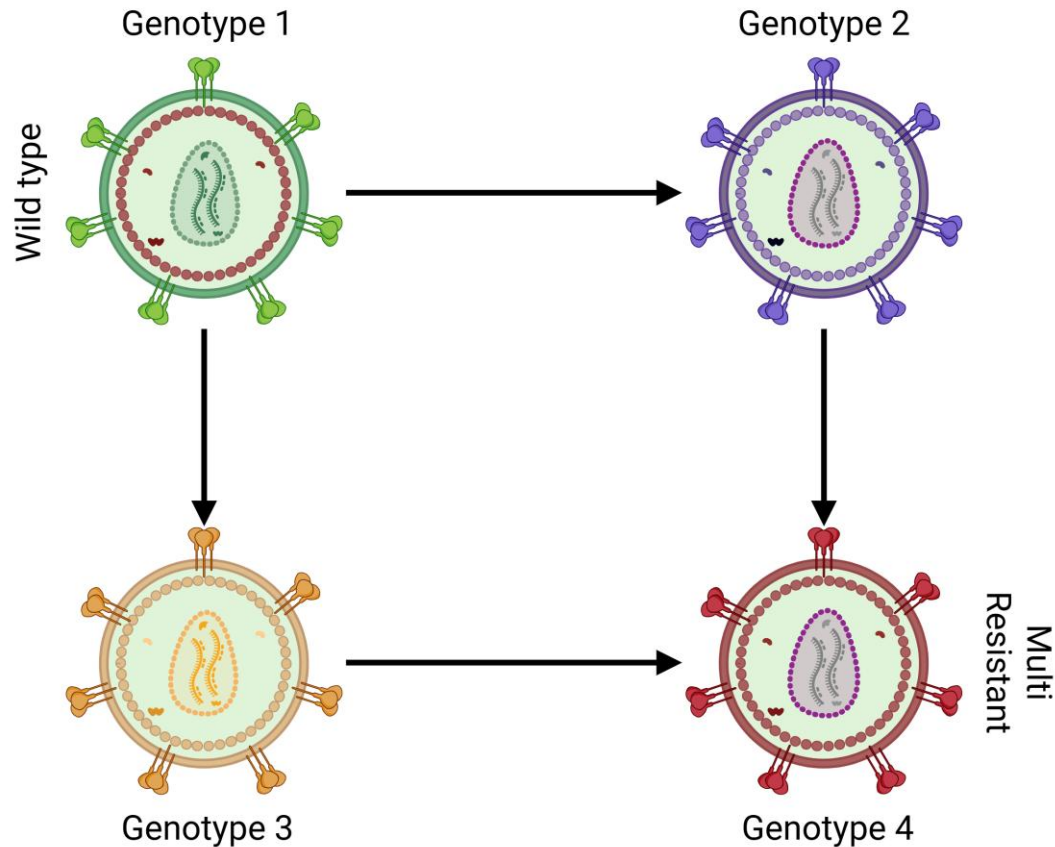


Figure 12. HIV mutation model scheme, 4 genotypes.

Taken from:

[10] Hernandez-Vargas, E. A., Colaneri, P., & Middleton, R. H. (2013). **Switching strategies to mitigate HIV mutation**. *IEEE Transactions on Control Systems Technology*, 22(4), 1623-1628.

HIV mutation

4 genotype – 4 parameter

Consider the HIV model (1) with 4 genotypes, the following change of coordinates

$$x = [T, M, T_1^*, M_1^*, V_1, T_2^*, M_2^*, V_2, T_3^*, M_3^*, V_3, T_4^*, M_4^*, V_4]/\lambda,$$

with,

$$y = [T, V_1, V_2, V_3, V_4]/\lambda,$$

transforms the model (1), see next slide. Under the assumption that $y_1 := T/\lambda$, $y_2 := V_1/\lambda$, $y_3 := V_2/\lambda$, $y_4 := V_3/\lambda$ and $y_5 := V_4/\lambda$ are measurable signals for all time, the system can be rewritten in the state affine form, Eq. (2). Therefore, matrices $A(y)$, $\beta(y)$, and $\varphi(y)$ can be computed for this scenario. In which the unknown parameters are stated as,

$$\theta = [K_{T,\sigma}^1, K_{T,\sigma}^2, K_{T,\sigma}^3, K_{T,\sigma}^4]^T,$$

HIV mutation

4 genotype – 4 parameter

$$\begin{aligned}\dot{x}_1 &= -\delta_T x_1 + \left(\frac{\lambda \rho_T y_1 y_{V_T}}{C_T + \lambda y_{V_T}} + \frac{S_T}{\lambda} \right) - \lambda y_1 y_2 K_{T,\sigma}^1 \\ &\quad - \lambda y_1 y_3 K_{T,\sigma}^2 - \lambda y_1 y_4 K_{T,\sigma}^3 - \lambda y_1 y_5 K_{T,\sigma}^4, \\ \dot{x}_2 &= \left(\frac{\lambda \rho_M y_{V_T}}{C_M + \lambda y_{V_T}} - \lambda y_2 K_{M,\sigma}^1 - \lambda y_3 K_{M,\sigma}^2 \dots \right. \\ &\quad \left. - \lambda y_4 K_{M,\sigma}^3 - \lambda y_5 K_{M,\sigma}^4 - \delta_M \right) x_2 + \frac{S_M}{\lambda}, \\ \dot{x}_3 &= -\delta_{T^*} x_3 + (1 - 2\mu) \lambda y_1 y_2 K_{T,\sigma}^1, \\ \dot{x}_4 &= (1 - 2\mu) \lambda y_2 K_{M,\sigma}^1 x_2 - \delta_{M^*} x_4, \\ \dot{x}_5 &= P_{T,\sigma}^1 x_3 + P_{M,\sigma}^1 x_4 - \delta_V x_5, \\ \dot{x}_6 &= -\delta_{T^*} x_6 + \mu \lambda y_1 y_2 K_{T,\sigma}^1 + (1 - \mu) \lambda y_1 y_3 K_{T,\sigma}^2, \\ \dot{x}_7 &= (\mu \lambda y_2 K_{M,\sigma}^1 + (1 - \mu) \lambda y_3 K_{M,\sigma}^2) x_2 - \delta_{M^*} x_7, \\ \dot{x}_8 &= P_{T,\sigma}^2 x_6 + P_{M,\sigma}^2 x_7 - \delta_V x_8, \\ \dot{x}_9 &= -\delta_{T^*} x_9 + \mu \lambda y_1 y_2 K_{T,\sigma}^1 + (1 - \mu) \lambda y_1 y_4 K_{T,\sigma}^3, \\ \dot{x}_{10} &= (\mu \lambda y_2 K_{M,\sigma}^1 + (1 - \mu) \lambda y_4 K_{M,\sigma}^3) x_2 - \delta_{M^*} x_{10}, \\ \dot{x}_{11} &= P_{T,\sigma}^3 x_9 + P_{M,\sigma}^3 x_{10} - \delta_V x_{11}, \\ \dot{x}_{12} &= -\delta_{T^*} x_{12} + \mu \lambda y_1 y_3 K_{T,\sigma}^2 + \mu \lambda y_1 y_4 K_{T,\sigma}^3 + \lambda y_1 y_5 K_{T,\sigma}^4, \\ \dot{x}_{13} &= (\mu \lambda y_3 K_{M,\sigma}^2 + \mu \lambda y_4 K_{M,\sigma}^3 + \lambda y_5 K_{M,\sigma}^4) x_2 - \delta_{M^*} x_{13}, \\ \dot{x}_{14} &= P_{T,\sigma}^4 x_{12} + P_{M,\sigma}^4 x_{13} - \delta_V x_{14},\end{aligned}$$

Results on Real-time parameter updating

Numerical Results

Scenario 2.1: 4 genotype – 4 parameter

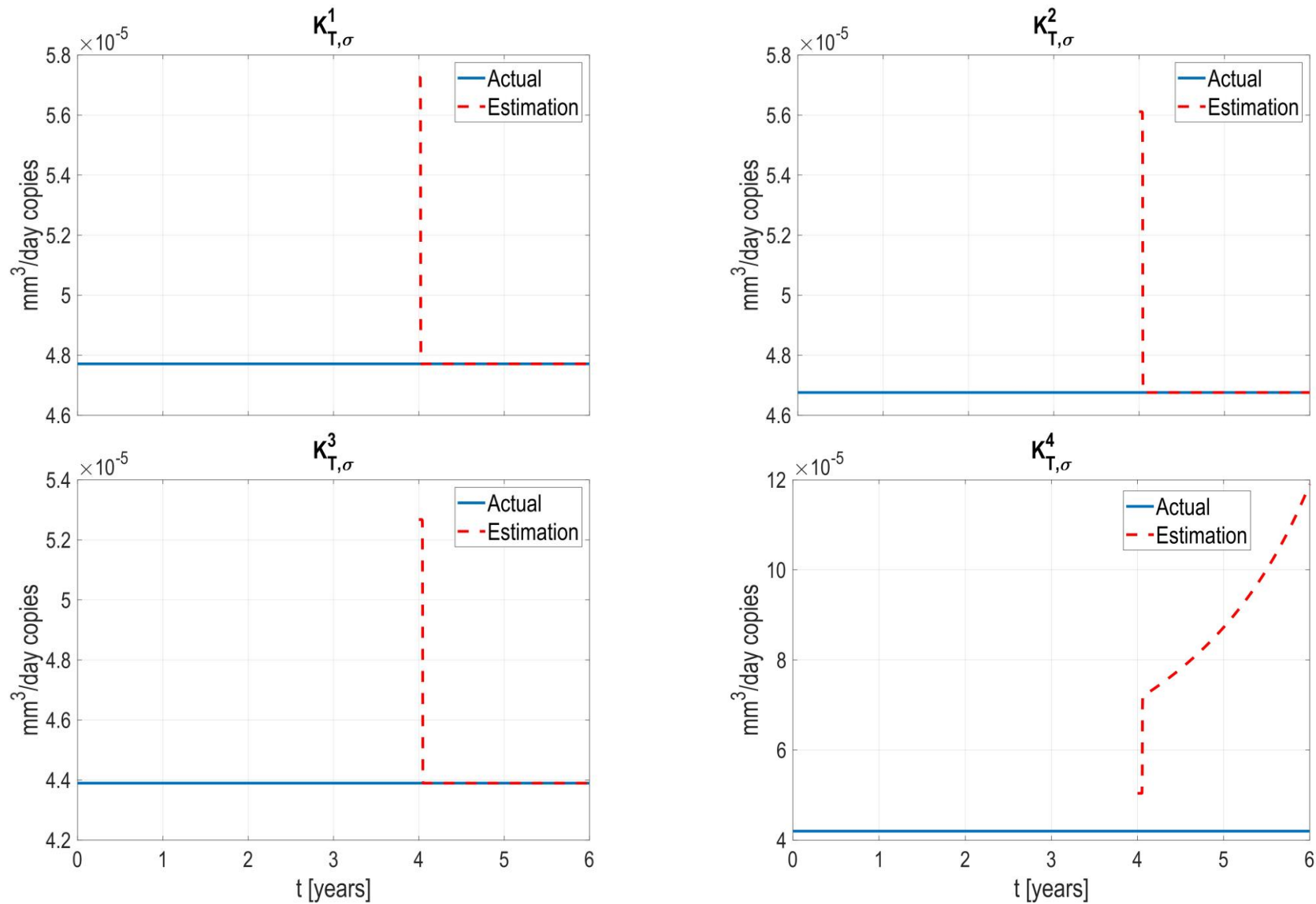
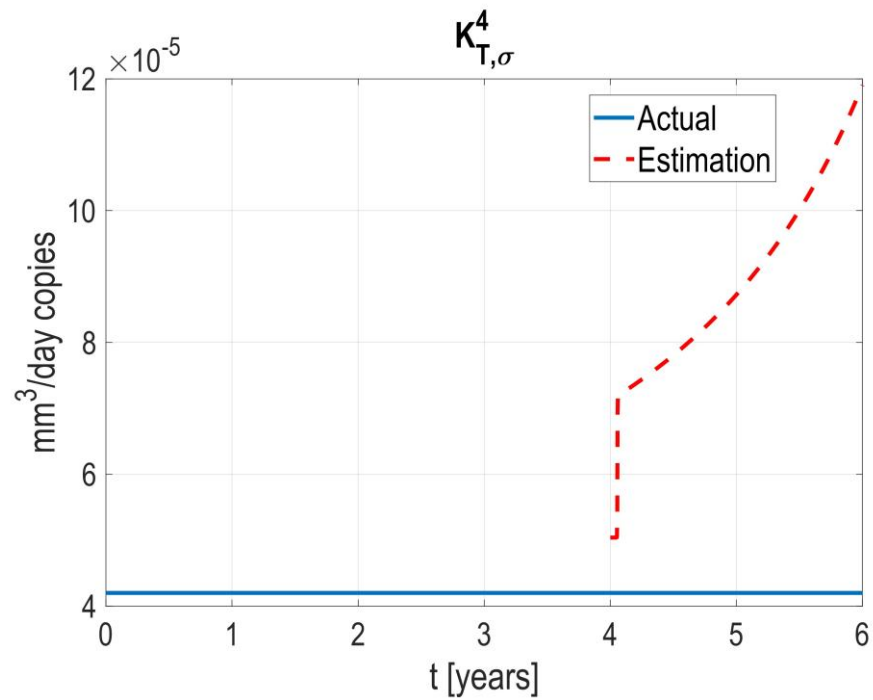


Figure 13. Parameter estimation of $K_{T,\sigma}^{i=1,2,3,4}$.

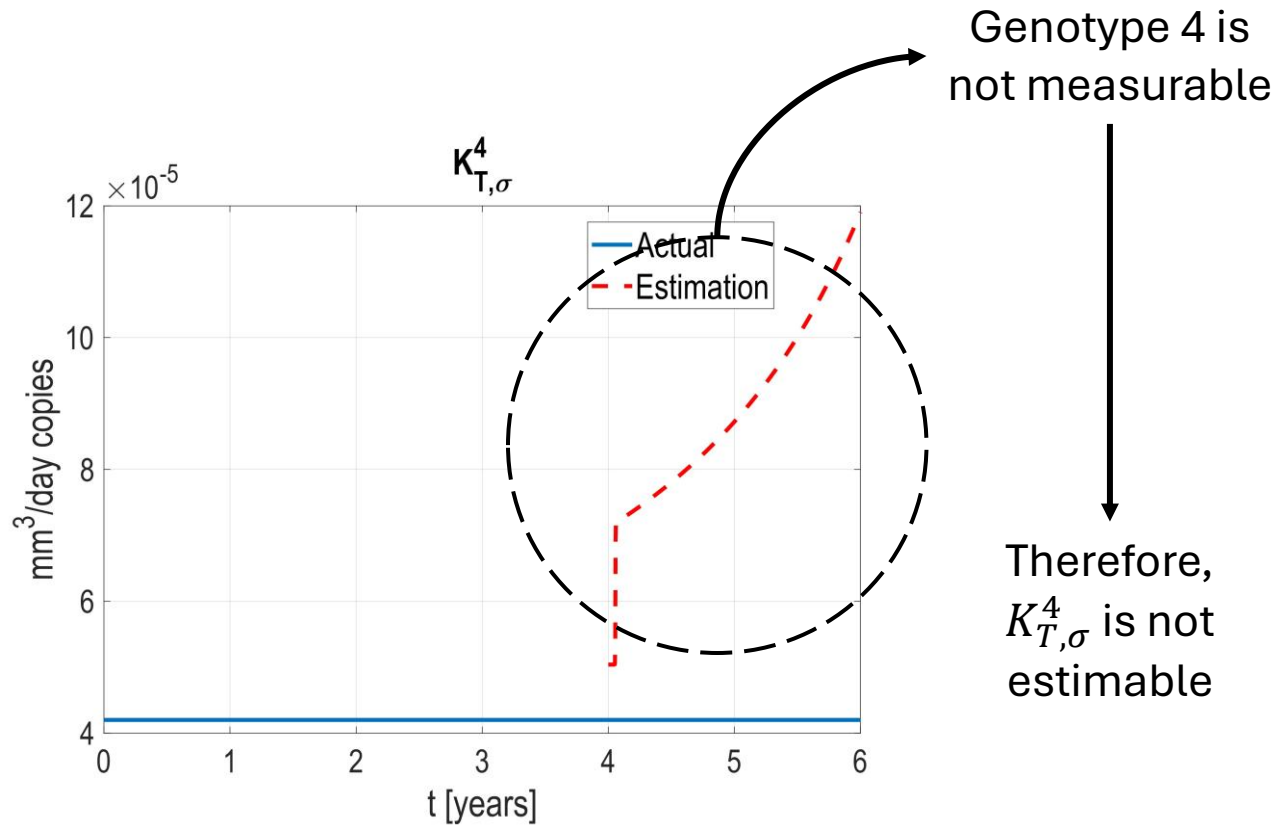
Numerical Results

Scenario 2.1: 4 genotype – 4 parameter



Numerical Results

Scenario 2.1: 4 genotype – 4 parameter



Numerical Results

Scenario 2.1: 4 genotype – 4 parameter

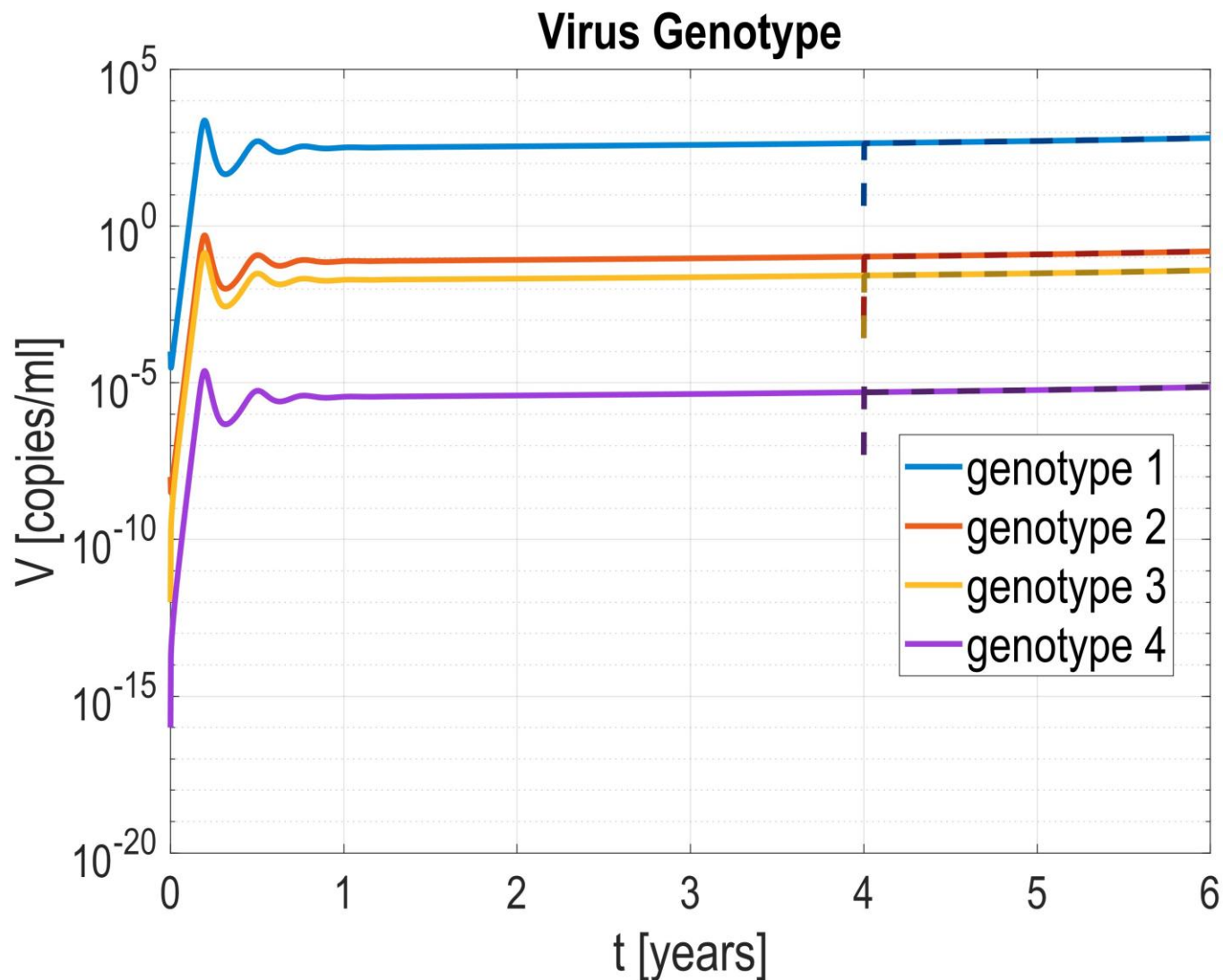


Figure 14. State estimation of each Virus genotype.

Numerical Results

Scenario 2.1: 4 genotype – 4 parameter

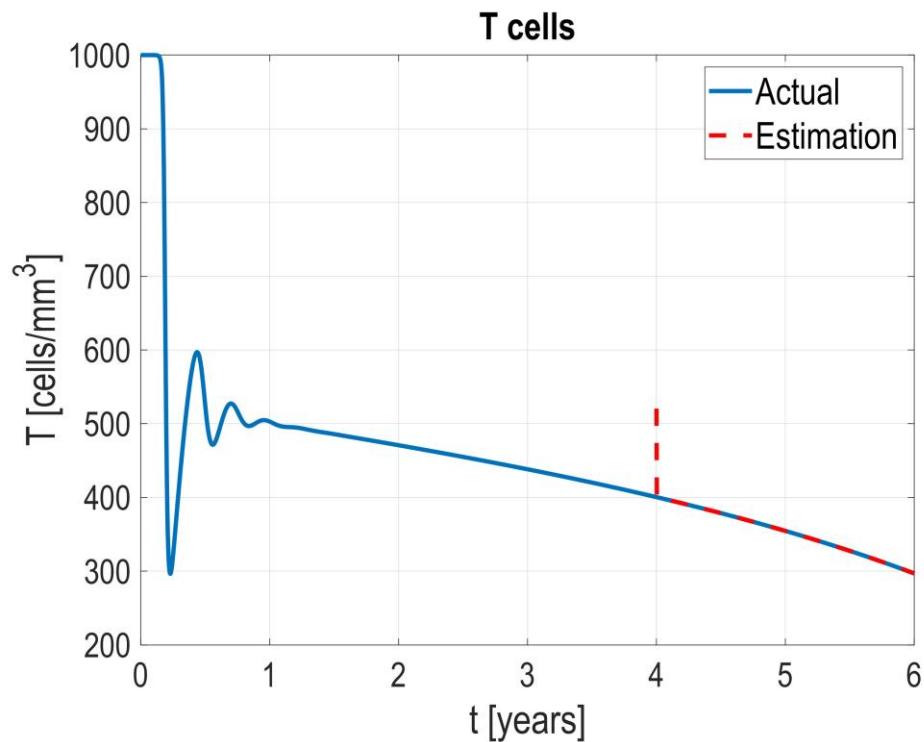


Figure 15. State estimation of CD4+ T cells.

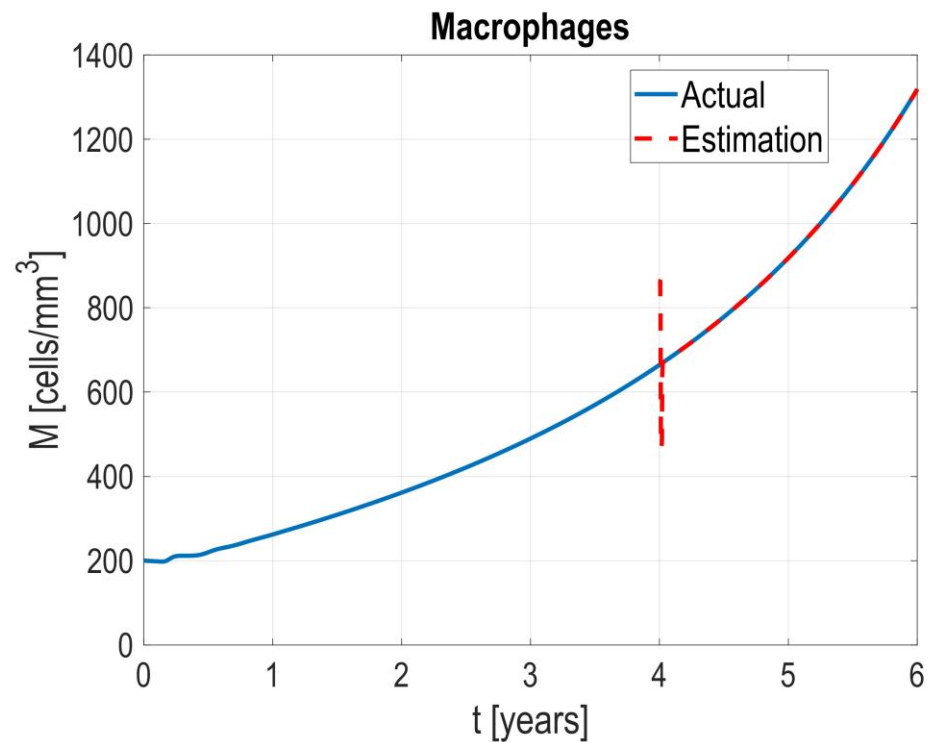


Figure 16. State estimation of Macrophages.

Numerical Results

Scenario 2.1: 4 genotype – 4 parameter

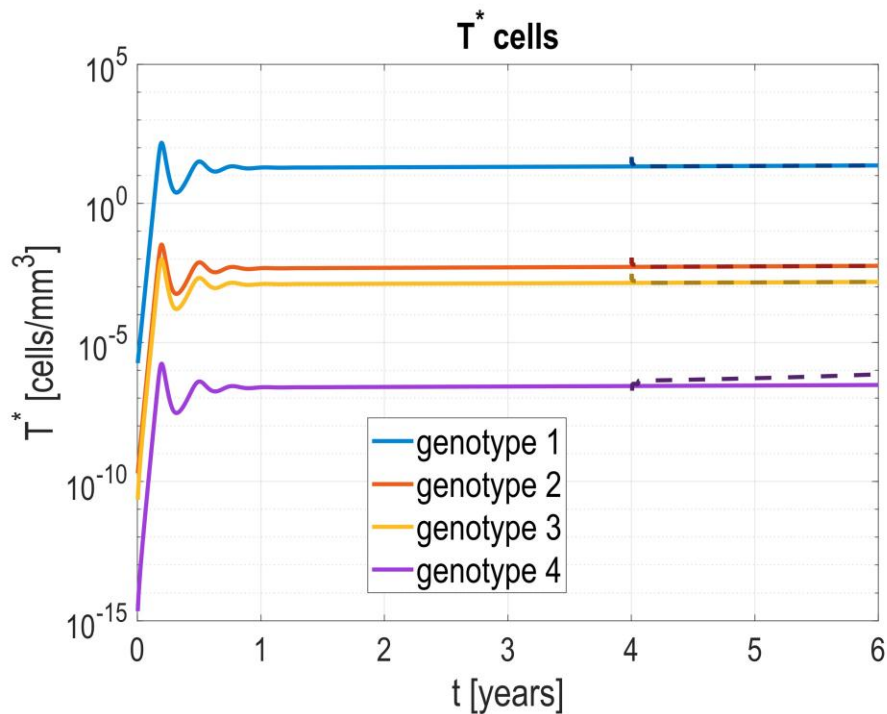


Figure 17. State estimation of Infected T cells, by each genotype.

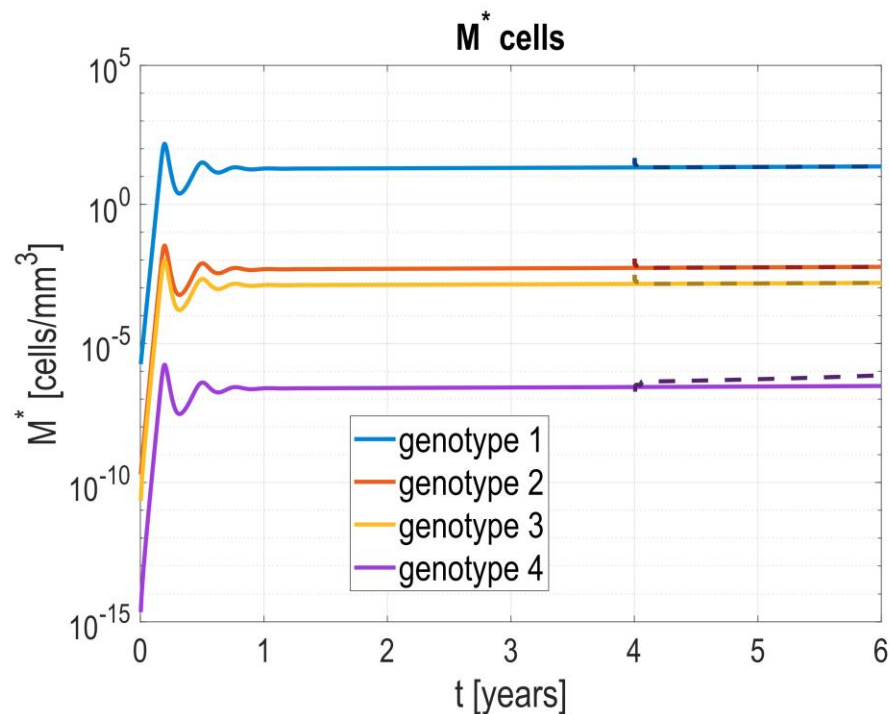


Figure 18. State estimation of Infected macrophages, by each genotype.

Numerical Results

Scenario 2.1: Switching Therapies.

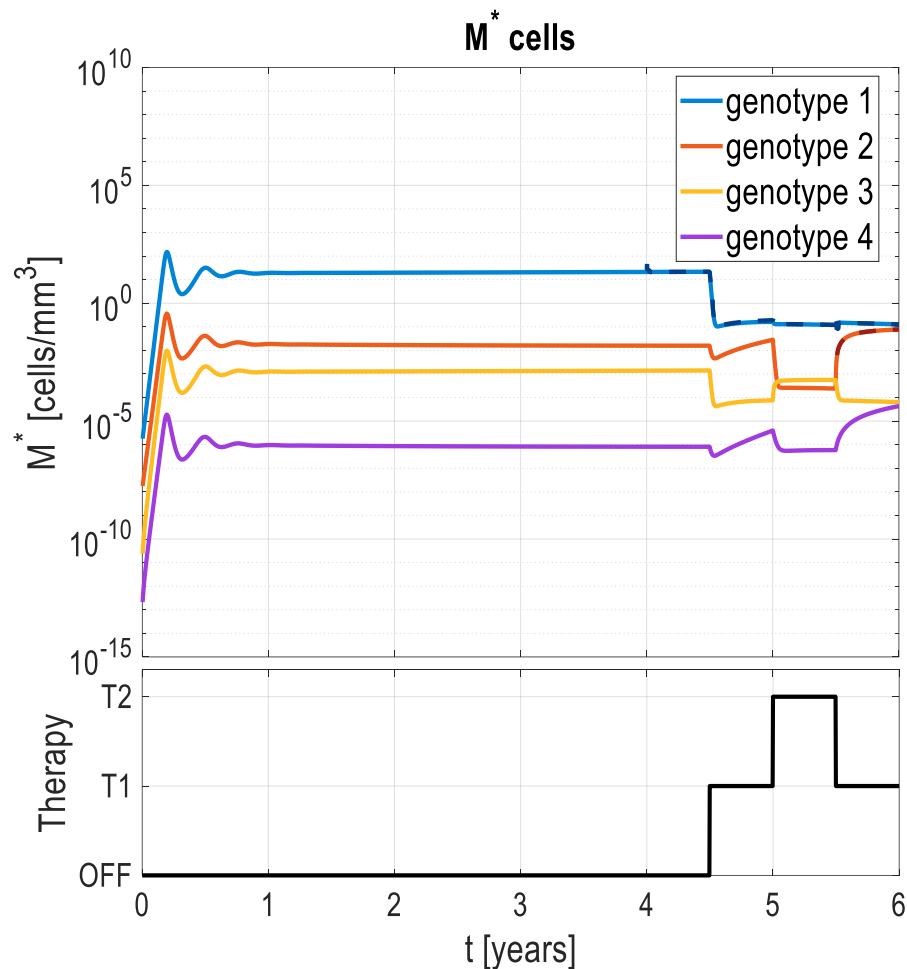
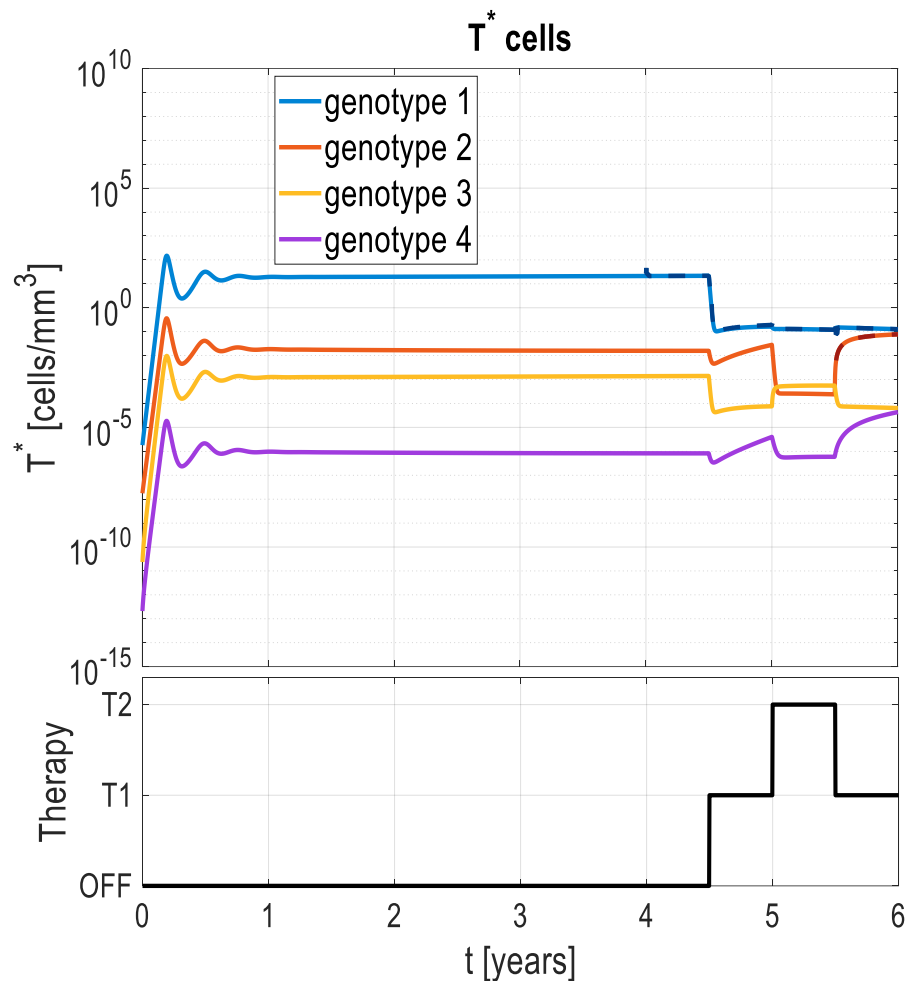


Figure 19. State estimation of Infected T cells, by each genotype.

Figure 20. State estimation of Infected macrophages, by each genotype.

On Hybrid Extended Kalman Filtering for Antimicrobial Resistance models

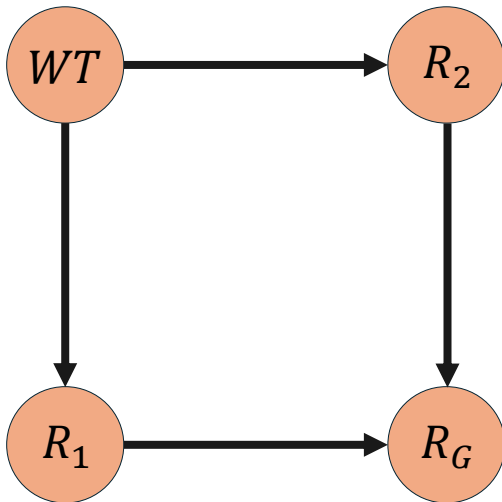
Antimicrobial Resistance model

$$\dot{x}_i(t) = \rho_{i,\sigma(t)} x_i(t) \left(1 - \frac{x_i(t)}{K}\right) - \delta x_i(t) + \mu \sum_{j \neq i}^{\eta} m_{i,j} x_j(t). \quad (4)$$

Antimicrobial Resistance model

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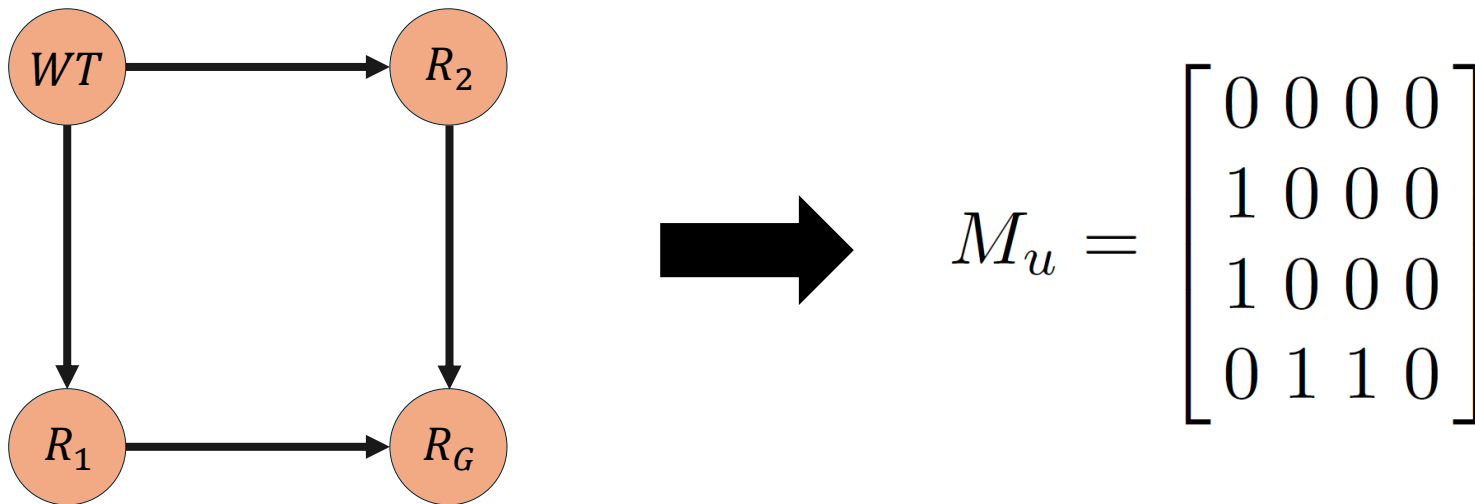
Mutation tree



Antimicrobial Resistance model

$$\dot{x}_i(t) = \rho_{i,\sigma(t)} x_i(t) \left(1 - \frac{x_i(t)}{K}\right) - \delta x_i(t) + \mu \sum_{j \neq i}^{\eta} m_{i,j} x_j(t). \quad (4)$$

Mutation tree



Hybrid Extended Kalman Filtering

Taken from:

[7] Simon, D. (2006). *Optimal state estimation: Kalman, H infinity, and nonlinear approaches*. John Wiley & Sons.

Hybrid Extended Kalman Filtering

1. Consider the system equations with continuous-time dynamics and discrete-time measurements are given as follows:

$$\begin{aligned}\dot{x} &= f(x, u, w, t) \\ y_k &= h_k(x_k, v_k) \\ w(t) &\sim (0, Q) \\ v_k &\sim (0, R_k)\end{aligned}\tag{5}$$

2. Initialize the filter as follows:

$$\begin{aligned}\hat{x}_0^+ &= E[x_0] \\ P_0^+ &= E[(x_0 - \hat{x}_0^+)(x_0 - \hat{x}_0^+)^T]\end{aligned}\tag{6}$$

Hybrid Extended Kalman Filtering

3. For $k = 1, 2, \dots$, perform the following.

- (a) Integrate the state estimate and its covariance from time $(k-1)^+$ to time k^- as follows:

$$\begin{aligned}\dot{\hat{x}} &= f(\hat{x}, u, 0, t) \\ \dot{P} &= AP + PA^T + LQL^T\end{aligned}\tag{7}$$

where A and L are defined by the following partial derivative matrices, evaluated at the current state estimate:

$$A = \left. \frac{\partial f}{\partial x} \right|_{\hat{x}} \quad L = \left. \frac{\partial f}{\partial w} \right|_{\hat{x}}\tag{8}$$

Hybrid Extended Kalman Filtering

3. For $k = 1, 2, \dots$, perform the following.

- (a) Integrate the state estimate and its covariance from time $(k-1)^+$ to time k^- as follows:

$$\begin{aligned}\dot{\hat{x}} &= f(\hat{x}, u, 0, t) \\ \dot{P} &= AP + PA^T + LQL^T\end{aligned}\tag{7}$$

where A and L are defined by the following partial derivative matrices, evaluated at the current state estimate:

$$A = \left. \frac{\partial f}{\partial x} \right|_{\hat{x}} \quad L = \left. \frac{\partial f}{\partial w} \right|_{\hat{x}}\tag{8}$$

We begin this integration process with $\hat{x} = \hat{x}_{k-1}^+$ and $P = P_{k-1}^+$. At the end of this integration we have $\hat{x} = \hat{x}_k^-$ and $P = P_k^-$.

- (b) At time k , incorporate the measurement y_k into the state estimate and estimation covariance as follows:

$$\begin{aligned}K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + M_k R_k M_k^T)^{-1} \\ \hat{x}_k^+ &= \hat{x}_k^- + K_k (y_k - h_k(\hat{x}_k^-, 0, t_k)) \\ P_k^+ &= (I - K_k H_k) P_k^- (I - K_k H_k)^T + K_k M_k R_k M_k^T K_k^T\end{aligned}\tag{9}$$

H_k and M_k are the partial derivatives of $h_k(x_k, v_k)$ with respect to x_k and v_k , and are both evaluated at $\hat{x}_k^-$.

Hybrid Extended Kalman Filtering

Joint State - parameter estimation

In order to estimate a parameter p , we first augment the state with the parameter to obtain an augmented state vector x' :

$$x'_k = \begin{bmatrix} x_k \\ p_k \end{bmatrix} \quad (10)$$

If p_k is constant then we model $p_{k+1} = p_k + w_{pk}$, where w_{pk} is a small artificial noise term that allows the Kalman filter to change its estimate of p_k .

Results on Hybrid EKF for AMR model

Numerical Results

Hybrid EKF for joint state-parameter estimation on AMR model

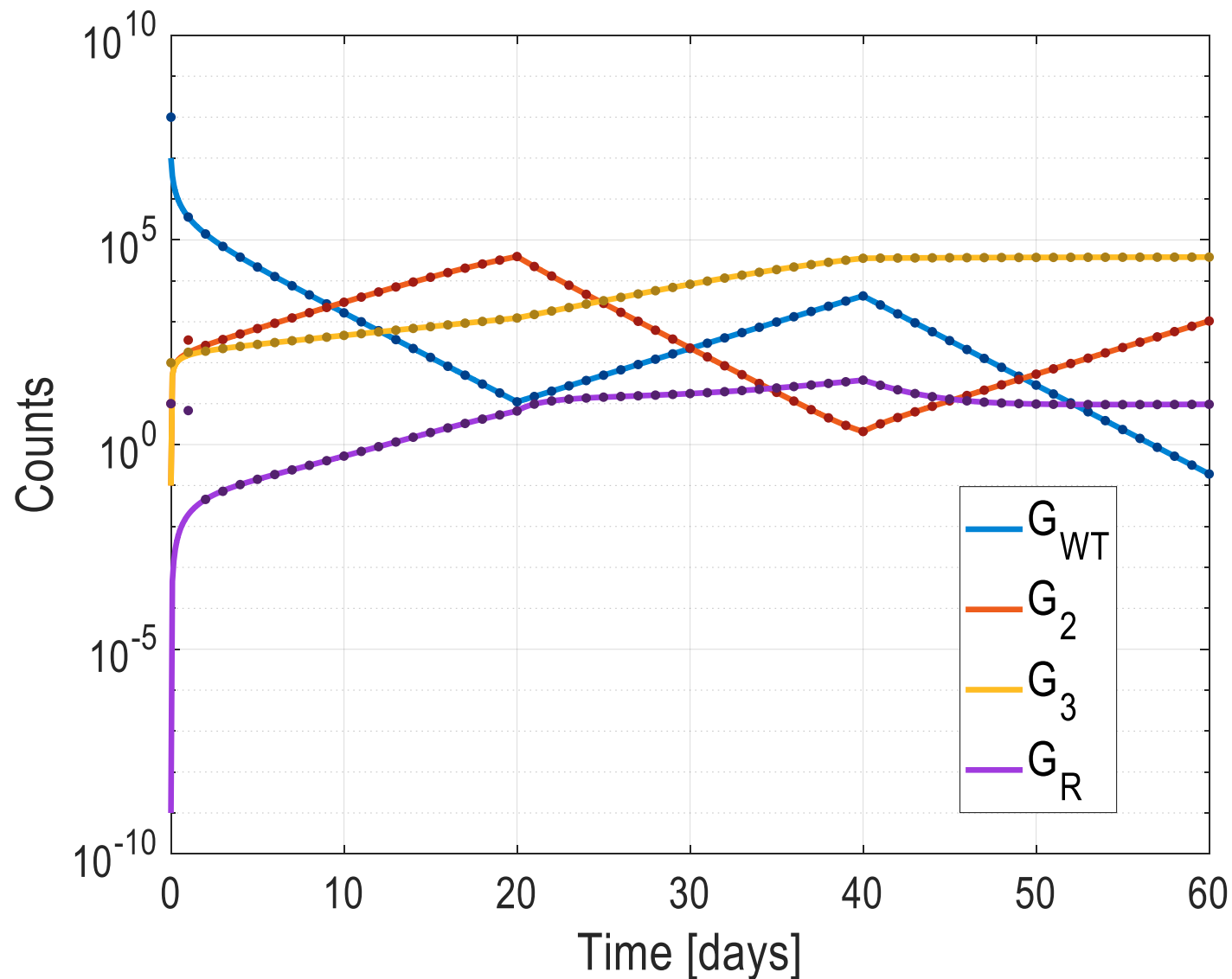


Figure 21. Temporal evolution of bacteria population.

Numerical Results

Hybrid EKF for joint state-parameter estimation on AMR model

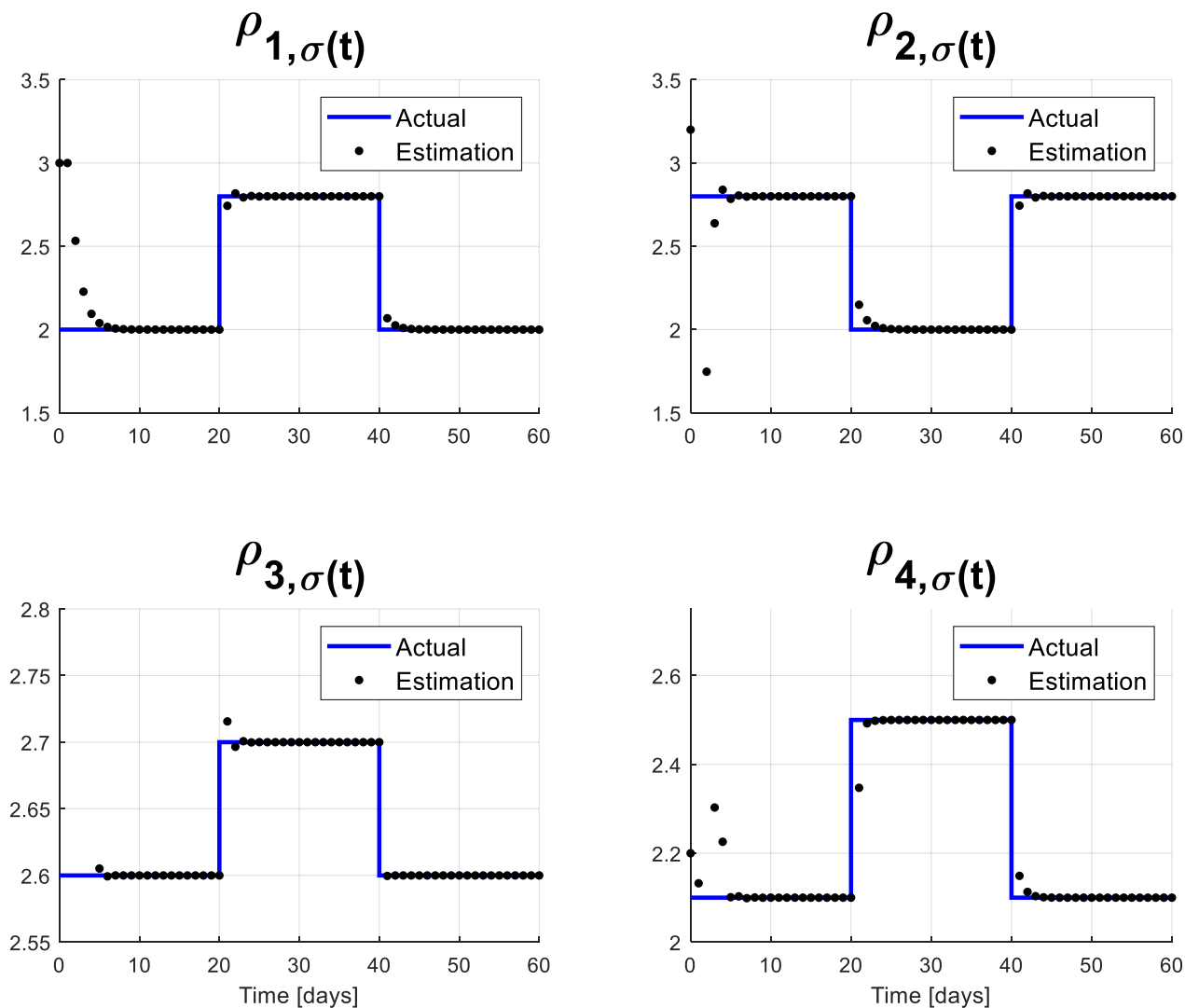


Figure 22. Parameter estimation of $\rho_{i,\sigma}$.

Numerical Results

Sampling time every 2 days

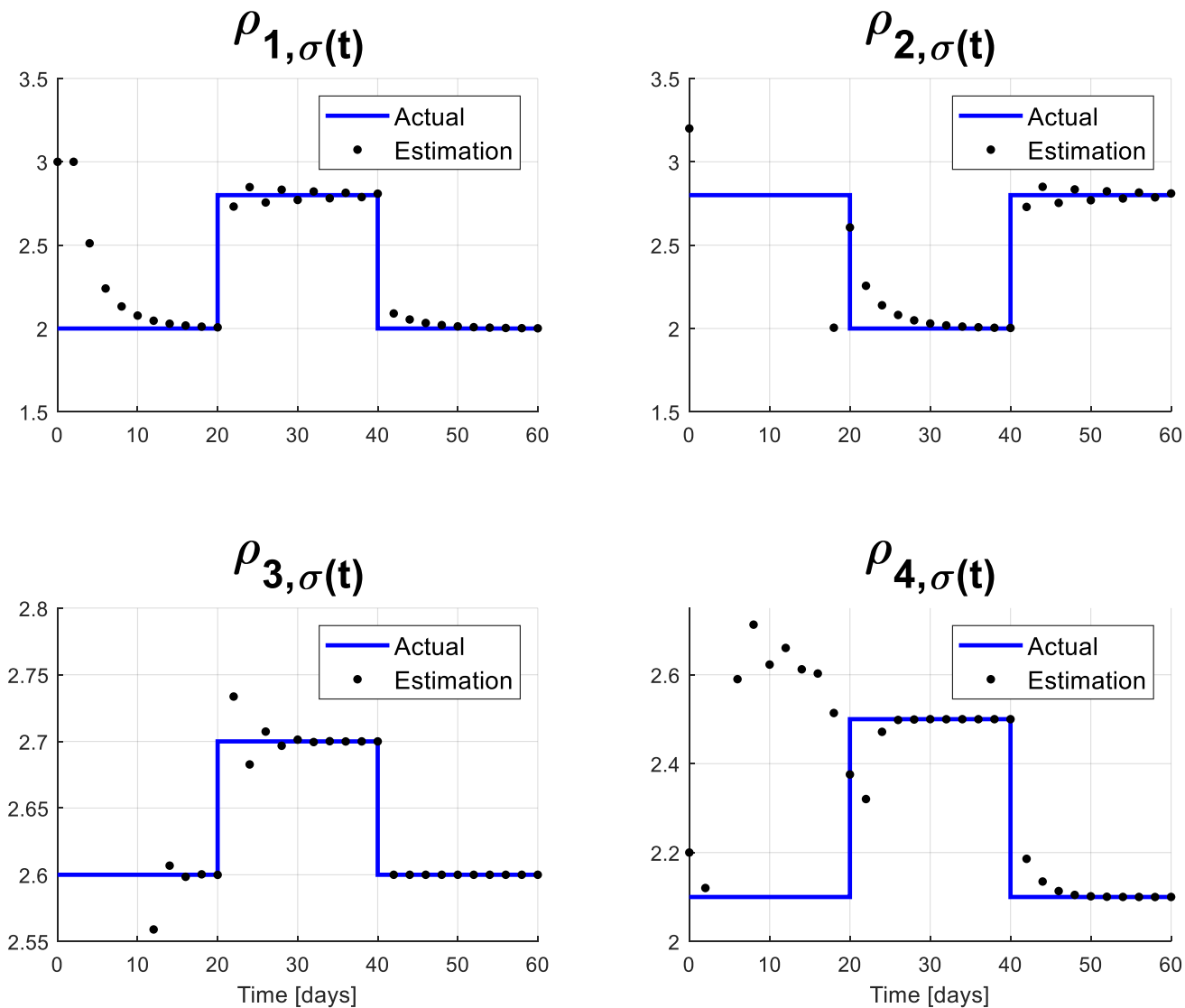


Figure 22. Parameter estimation of $\rho_{i,\sigma}$.

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of Infectious Diseases**
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Our research group seeks new avenues for holistic understanding against infections at the meeting point of mathematics, machine learning, and infection biology. By integrating multi-scale experimental data, modeling, and the power of computational simulations we aim to:

COVID-19 Research

Tweets by @systmedicine

Esteban A. Hernandez-Vargas Retweeted
Jesus Alvarez
@Jgac79
Replying to @systmedicine



Contact us at:

esteban@systemsmedicine.de

**Thanks for attending,
questions ?**

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Conclusions

- Our model of HIV infection include sufficient mechanistic details in a way that parameters can be periodically updated with the new measurable data.
- The real-time parameter updating algorithm ensure accurate predictions for the HIV infection.
- Adaptive observers are originally formulating in continuous time; therefore, it suffer when considering sampling scenarios.

Conclusions

- 3
- 2
- 1

References

- [1] Wright, L., & Davidson, S. (2020). How to tell the difference between a model and a digital twin. AMSES, 7, 1-13.
- [2] Hernandez-Vargas, E. A., & Middleton, R. H. (2013). Modeling the three stages in HIV infection. Journal of theoretical biology, 320, 33-40.
- [3] Rodríguez, A., et al. An adaptive observer for operation monitoring of anaerobic digestion wastewater treatment. J. Chem. Eng. 269 (2015): 186-193.
- [4] Hernandez-Vargas, E. A. (2019). Modeling and control of infectious diseases in the host: with MATLAB and R. Academic Press.

Performance Results[†]

Table 1. Scenario 1.1

	$K_{T,\sigma}^1$	Virus	CD4+ T cells	Macrophages	Infected T cells	Infected Macro.
RMSLE	$1.662e - 06$	0.1834	0.0109	0.0520	0.0447	0.0021

Table 2. Scenario 1.2

	$K_{T,\sigma}^1$	Virus	CD4+ T cells	Macrophages	Infected T cells	Infected Macro.
RMSLE	$1.022e - 07$	0.0030	$1.7920e - 04$	0.0081	$6.2431e - 04$	$6.3266e - 04$

[†] Both scenarios consider $K_{T,\sigma}$ is an unknown parameter and only V and T are measurable variables.